

SYSTEMIC RISK IN FINANCIAL NETWORKS

REVISITED: DEBT DILUTION AS A BACKDOOR

BAIL-IN*

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September 30, 2026

Abstract

We develop a model of interbank networks with random liquidity shocks. Networks of dilutable debt—e.g., long-term, unsecured—facilitate efficient liquidity transfers: Shocked banks pledge interbank claims as collateral for new senior debt, diluting existing debt. Unlike with non-dilutable debt, indebtedness and connectedness are sources of stability, not fragility. Dilution is thus a “backdoor bail-in” that reallocates losses absent a resolution authority, trigger security, or ex post renegotiation. We uncover a class of networks, “exponential networks,” that implement optimal contingent transfers via plain debt. Yet exponential networks are not pairwise stable, whereas some core–periphery networks are, rationalizing observed interbank structures and their under-insurance against crises.

*For helpful comments, thanks to Franklin Allen, Vladimir Asriyan, Ana Babus, Bruno Biais, Philip Bond, Agostino Capponi, Tetiana Davydiuk, Doug Diamond, Darrell Duffie, Phil Dybvig, Douglas Gale, Todd Gormley, Denis Gromb, Zhiguo He, Florian Heider, Laurie Hodrick, Sebastian Infante, Dalida Kadyrzhanova, Martin Oehmke, Mina Lee, Francesco Palazzo, Uday Rajan, Adriano Rampini, Victoria Vanasco, Laura Veldkamp, Thao Vuong, Chaojun Wang, David Webb, Alexander Zentefis, Jeff Zwiebel, and audiences at the 2018 AFA, Banca d’Italia, EIEF, the 2017 ECWFC Meeting, the 2017 Summer Meeting of the FTG, the 2017 FTG Summer School, the FRA, the 2017 Summer Symposium at Gerzensee, the 2018 Maryland Junior Finance Conference, Washington University in St. Louis, and the 2017 Yale Junior Finance conference. Thanks to Shunsuke Matsuno and Dayou Xi for research assistance. This paper replaces Donaldson and Piacentino (2018).

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1 Introduction

Interbank debts are viewed as a threat to financial stability, partly due to a theory literature that shows how a shock to one bank can transmit to its counterparties; as a result, tightly interconnected networks of debt are “robust yet fragile,” absorbing everyday shocks but amplifying extraordinary ones (see, notably, Acemoglu, Ozdaglar, and Tahbaz-Salehi (2015), hereinafter AOT, Allen and Gale (2000), and Eisenberg and Noe (2001)). This fragility has motivated policies that shift a distressed bank’s losses onto its healthy counterparties, aiming to reallocate liquidity and contain distress—statutory bail-ins, contingent convertibles, and regulator-coordinated private recapitalizations.¹ Such policies generally require something beyond ordinary debt: a resolution authority, an explicit contingent security, or an ex post agreement among counterparties.

The corporate finance literature points to a different possibility: Ordinary debt may itself implement such transfers. Debt that can be diluted—so new senior debt can be issued before existing debt is paid—allows a distressed debtor to raise new debt at the expense of old, thereby avoiding inefficient liquidation.² The mechanism, which relies on gross interbank claims, suggests a downside of netting out.^{3,4} Can gross positions be a source of insurance rather than contagion in the financial system? If so, would netting them out destroy coinsurance?

¹Examples include statutory bail-in powers (Bolton and Oehmke (2019) and Kanik (2022)); contingent convertible securities (Flannery (2014) and Sundaresan and Wang (2015)); private recapitalizations and rescue arrangements (Leitner (2005), Rogers and Veraart (2013), Kanik (2020), and Bernard, Capponi, and Stiglitz (2022)); and public interventions targeted to financial networks (Erol (2019), Capponi, Corell, and Stiglitz (2022) and Jackson and Pernoud (2024)).

²See, among others, Stulz and Johnson (1985), Diamond (1993), Hart and Moore (1995), and Donaldson, Gromb, and Piacentino (2020, 2025); see also Donaldson and Piacentino (2026) for a survey of the literature on collateral, dilution, and priority.

³Existing evidence suggests that a substantial fraction of interbank debt has maturity long enough for dilution to be relevant. E.g., several papers using German data report average maturities exceeding one year, with overnight debt accounting for only about 10% of exposures (e.g., Bluhm et al. (2016), Craig and Ma (2022), Craig and Von Peter (2014), Gabrieli and Georg (2014), Upper and Worms (2004)). Kuo et al. (2014) document the scarcity of U.S. maturity data and infer from payment records that roughly one quarter of interbank debt has term maturity.

⁴See, e.g., Allen and Gale (2000), Acemoglu et al. (2015), Allen et al. (2012), Kusnetsov and Veraart (2019), He and Li (2022) and Shu (2025).

We develop a model of a financial system with dilutable interbank debt that suggests that the answer to both questions is yes: Gross debt positions insure banks against liquidity shocks, so netting them out destroys valuable risk sharing. In diametric contrast to what happens in networks of non-dilutable debt, high indebtedness and connectedness become sources of stability: More debt provides collateral against which distressed banks can borrow, and denser connections allow liquidity to flow from banks that have it to banks that need it.

High indebtedness alone does not guarantee efficiency. The distribution of debt across banks matters too. However, we identify a heretofore unexplored class of networks, which we call “exponential networks,” that implement the efficient allocation of liquidity for every realization of shocks. They are robust and never fragile, implementing a “backdoor bail-in”: Ordinary debt achieves the desired contingent transfers without a resolution authority, contingent security, or ex post negotiation.

That does not mean there is no role for outside intervention. Exponential networks are unlikely to arise via free contracting (in the jargon, they are not pairwise stable). Core-periphery networks like those observed in practice⁵ are more likely to arise but under-insure against crises. Our analysis therefore suggests that, when the loss-absorbing capacity comes from within the financial system through dilution, some banks being “too big to fail” can improve rather than undermine systemic stability.

Prima facie, the model tells a policy maker to impose an exponential network of interbank debts. We would not be so literal given how abstract the model is. But two lessons seem robust: Symmetric networks cannot insure banks against large shocks and networks of non-dilutable debt cannot insure them at all. Policies that compress gross exposures, cap the largest ones, or curtail the superpriority of new secured claims may therefore trade real insurance for the appearance of safety.⁶

To study debt dilution in the financial system, we embed the liquidity-risk framework of

⁵See, e.g., Craig and Von Peter (2014), Gabrieli and Georg (2014), and Craig and Ma (2022).

⁶Examples include EMIR portfolio-compression requirements, Basel capital deductions for banks’ holdings of other G-SIBs’ TLAC instruments and large-exposure limits, and U.S. QFC stay and clean-holding-company rules.

Holmström and Tirole (1998) in a network. Banks have imperfectly pledgeable long-term assets and may suffer liquidity shocks before those assets mature. They also owe debts to and from one another, which can be seen as cross-holdings of loans or bonds. In the baseline, interbank debt matures after the shock, so it can be diluted with new senior debt, representing, e.g., repos, which have superpriority in bankruptcy (Duffie and Skeel (2012)). In a benchmark, it matures at the time of the shock, too soon to be diluted to raise liquidity.

The benchmark turns out to be isomorphic to AOT’s model (Lemma 1), allowing us to import their measures of connectedness and recover (and sometimes strengthen) their results: With non-dilutable debt, netting improves stability, connectedness reduces it, and there is a “default radius” around a shocked bank (Lemmata 2–5).

Our first set of results shows how each of these conclusions reverses when debt can be diluted. In that case, netting undermines stability (Proposition 2), connectedness enhances it (Propositions 3 and 5), and there is a “salvation radius” around a healthy bank (Proposition 4). Behind the reversal is a change in what the network transmits: Rather than spreading liquidity shortages from shocked banks, it spreads liquidity surpluses from healthy ones.

High connectedness and indebtedness are not enough for (constrained) efficiency. The distribution of links matters too. For instance, symmetric networks tend to be inefficient because they allocate liquidity equally to all shocked banks, “wasting” it on banks that cannot be saved.

In our second set of results, we explore what we call exponential networks. We show that these networks, which are tightly connected but asymmetric, implement an efficient priority rule, allocating all free liquidity to the largest set of banks that can be saved and writing off other shocked banks entirely (Proposition 6): An optimal state-contingent transfer scheme built entirely from plain, non-contingent debt. That shows how debt dilution extends the contingencies that ordinary debt can implement beyond what can be done via default alone (Allen and Gale (1998), Dubey, Geanakoplos, and Shubik (1988), Zame (1993)).⁷

⁷Since dilution requires new debt to be issued before outstanding debt matures, the mechanism also points to a different way that maturity can substitute for explicit state contingency, complementing Angeletos (2002)

Our third set of results pertains to a version of endogenous network formation (pairwise stability of the face values between linked banks⁸). The exponential network is—alas!—not pairwise stable. Versions of core–periphery networks are. Together these results suggest our model can explain why core–periphery structures can persist and how banks underinsure against crises—the networks banks sustain privately can be pairwise stable but fail to provide the socially optimal allocation of liquidity.

For our fourth set of results, we extend the model, allowing banks to differ in their assets and liquidity shocks. We show that the planner’s problem is equivalent to what is known as the knapsack problem (Proposition 13) and that the exponential network implements the greedy algorithm, a method used to solve it approximately (Proposition 14). Here the network’s structure fixes the order in which banks are saved: The same ranking applies in every state, so the network is efficient if the planner’s priorities are not state contingent as is the case, e.g., if banks have identical liquidity needs (Corollary 2); otherwise it admits explicit efficiency guarantees (Corollary 3). Exponential networks are robust not only to which banks are shocked, but also to heterogeneity across banks.

We analyze three other extensions, the first two of which qualify our rosy view of large long-term debts (Section 8). In one, liquidation is efficient, so not every shocked bank should necessarily be rescued; in the other, any default is socially costly, not only those that induce inefficient liquidation. In both cases, we characterize debt levels that implement the efficient outcome, albeit only for specific network structures. The message is the same as in each: As in the baseline, debts should be large enough to provide liquidity insurance when liquidation is inefficient, but, unlike in the baseline, not so large that they induce inefficiency (be it excessive continuation or excessive default).

In our final extension, we let banks choose their asset pledgeability at a cost. We find, albeit in a two-bank example, that cross-holdings of dilutable debt can support efficient pledgeability choices: As long as pledgeability is worth having in autarky, every socially

and Gale (1990).

⁸So netting out is always allowed, but forming completely new links is not.

optimal profile of choices is an equilibrium. The reason is that a bank’s liquidity reaches its counterparty only in the states in which the bank is not shocked itself, so a bank that cuts back loses its own protection first. As elsewhere, interbank debts cannot be too low: It is the claim on a counterparty that makes its liquidity available.

The results offer a new perspective on inside liquidity. In Holmström and Tirole (1998, 2011), there is a continuum of agents with idiosyncratic shocks. They show that everyone holding a share of a diversified market index implements efficiency. That is not so in our setting with a finite number of agents, as the cross-section of shocks is a source of risk. In this case, we show that efficiency requires triage among shocked banks: Saving some and writing off others. That could be implemented by pooling assets in a portfolio, à la Holmström and Tirole, with triage implemented by an active fund manager. But intermediation is costly and could introduce frictions of its own. Our answer requires no intermediary at all. Bilateral debts encode the priority ordering *ex ante* in the profile of face values. Inside liquidity can thus be reallocated among a few large, interconnected institutions with no fund, no manager, and no contingent claims.

Overall, we provide a new way to think about financial stability and bank resolution. Much of the literature studies mechanisms to reallocate losses once a bank is in distress; we show that ordinary debt can do the same thing. Building on the corporate-finance insight that debt dilution provides liquidity, we show that interbank debt acts as a private bail-in in our model, reallocating losses through contractual priority rather than regulatory intervention. As a result, gross interbank debt need not propagate illiquidity; it can propagate liquidity too. As such, netting could reduce risk sharing and, moreover, large, highly interconnected banks stabilize rather than destabilize the financial system. Debt dilution thus reverses the financial stability role of financial networks.⁹

The bail-in mechanisms in the literature tend to require that someone do something or hold something beyond plain debt: Statutory bail-ins and public interventions require

⁹Surveys of the financial networks literature include Allen, Babus, and Carletti (2009), Allen and Walther (2021), Glasserman and Young (2016), and Jackson and Pernoud (2021).

an authority to act ex post (Bolton and Oehmke (2019), Erol (2019), Kanik (2022), Capponi, Corell, and Stiglitz (2022), Jackson and Pernoud (2024)); contingent capital requires an explicit contingent security, the trigger of which must be designed with care (Flannery (2014), Sundaresan and Wang (2015)); private recapitalizations require creditors to be willing to inject liquidity or accept haircuts ex post (Leitner (2005), Rogers and Veraart (2013), Kanik (2020), Bernard, Capponi, and Stiglitz (2022)). Dilution requires none of these: It is private—no authority stands between distress and transfer; it is written in plain debt—the contingency comes from the shocked bank’s borrowing, not from a contractual trigger, so there is no trigger to design and no trigger-induced multiplicity (our payment equilibrium is generically unique; Proposition 1); and it embeds commitment for free—banks cannot refuse to be diluted.

Still, the mechanism has its own limitations: It is only approximately efficient when banks are heterogeneous (Section 7); it relies on high debts, which can induce excessive continuation or costly default (Section 8); because the right network is unlikely to arise via free contracting (Section 6), it calls for regulation of network structure; and it requires the debt market to be open for shocked banks. But each limitation comes with a mitigant: The approximation admits explicit efficiency guarantees (Corollaries 2 and 3); debt levels can be calibrated to trade insurance off against excessive continuation and default; the regulation required is ex ante—of the network, not of banks in distress—and so creates no expectation of rescue to distort incentives; and the required market is for safe, secured debt, mitigating concerns of information-based freezes.

The rest of the paper proceeds as follows. Section 2 presents the model. Section 3 considers the benchmark without dilution. Section 4 states the qualitative properties of networks when debt can be diluted. Section 5 analyzes the exponential network. Section 6 asks which networks arise from decentralized contracting. Section 7 allows banks to be heterogeneous. Section 8 analyzes further extensions. Section 9 concludes. The Appendix contains all proofs, a worked example on the ring network, and a table of notations.

2 Model

We consider a model with two dates $t \in \{1, 2\}$ and $N \geq 2$ agents $B_1, \dots, B_N$, which we refer to as “banks.” Each has the risk of a liquidity shock ℓ at Date 1 and a long-term asset worth y if held until Date 2 but with value only $\theta y < y$ to outsiders (at either date).^{10,11} We refer to θy as the “pledgeable part” of a bank’s (non financial) asset and to $(1 - \theta)y$ as the “non-pledgeable” part.¹² There is universal risk neutrality and no discounting. All consumption is at Date 2.

Banks also have debts both to and from other banks—“interbank liabilities” and “interbank claims”—due at Date 2. The (gross) face value of B_i ’s liability to B_j is denoted by $F_{i \rightarrow j} \geq 0$, of all its liabilities to other banks by $F_{i \rightrightarrows} := \sum_{j \neq i} F_{i \rightarrow j}$, and of its claims on other banks by $F_{i \leftarrow} := \sum_{j \neq i} F_{j \rightarrow i}$; $\mathbf{F}_{\rightrightarrows}$ denotes the vector with i th element $F_{i \rightrightarrows}$. The matrix $\mathbf{F} := [F_{i \rightarrow j}]_{ij}$ defines the interbank network.^{13,14} Following AOT, we assume throughout that $F_{i \leftarrow} = F_{i \rightrightarrows}$ for all i , i.e. banks have zero net interbank positions.¹⁵ That is a

¹⁰The liquidity shock can be interpreted as a bank run. Under that interpretation, banks that are *not* run should still owe deposits at Date 2—senior claims that, unlike interbank debts, cannot be diluted. Adding such deposits, of size $\tau \ell$ with $\tau \ell < \theta y$, gives back the baseline model with transformed parameters: Replace y , ℓ , and θ with $y^\tau := y - \tau \ell$, $\ell^\tau := (1 - \tau)\ell$, and $\theta^\tau := (\theta y - \tau \ell)/(y - \tau \ell)$, and all results apply as stated. Economically, Date-2 deposits encumber part of a healthy bank’s pledgeable assets, reducing the liquidity it can supply to others without changing the mechanism. (If $\tau \ell \geq \theta y$, healthy banks have no unencumbered liquidity at all and the mechanism shuts down.) We thank a referee for suggesting this extension. (Cash holdings are the mirror image; see footnote 11.)

¹¹The model can also be interpreted as including cash holdings: If cash is perfectly pledgeable, then a bank endowed with cash $y^\$$ and productive assets $y^{-\$}$ with pledgeable fraction $\theta^{-\$}$ is just the baseline bank relabelled, with $y = y^{-\$} + y^\$$ and $\theta y = \theta^{-\$} y^{-\$} + y^\$$, and all results apply verbatim. Cash is the mirror image of the deposits in footnote 10: Deposits encumber a bank’s pledgeable assets; cash augments them, without changing the mechanism. We thank a referee for suggesting this interpretation. Section 8.3 lets banks choose θ , and hence, under this interpretation, how much cash to hold.

¹²The formulation can be interpreted literally in terms of private benefits or cash diversion; for micro-foundations in terms of other agency problems, see, e.g., DeMarzo and Fishman (2007) and Donaldson, Gromb, and Piacentino (2021).

¹³For most of the analysis, we focus on what debts in place do, staying agnostic about where they come from, be it to allocate liquidity, to incentivize monitoring, or to make markets; in Section 6, however, we explore endogenous debts.

¹⁴The interbank positions are modeled as plain debts. But many of our results are more general: Relying only on dilutability, they should also apply to cross-holdings of equity as in Elliott, Golub, and Jackson (2014) (see Section 4). Other results, like those on network design (Section 5), are specific to debt. Indeed, we see the strength of these results as implementing state-contingent transfers with non-contingent contracts, i.e. with debt.

¹⁵The zero-net assumption simplifies much of the analysis and is necessary for Definition 5. But the proofs of many results do not use it, notably those of Propositions 1, 2, and 3 and, under strong connectedness

reasonable approximation of reality, as gross interbank positions are often an order of magnitude larger than net.¹⁶ Whereas the face values $F_{i \rightarrow j}$ of interbank debt are exogenous for now (see, however, Section 6), the associated repayments are to be determined in equilibrium. B_i 's repayment to B_j is denoted by $R_{i \rightarrow j}$, its total repayment to all other banks by $R_{i \rightrightarrows} := \sum_{j \neq i} R_{i \rightarrow j}$, and its total repayment from all other banks by $R_{i \leftrightsquigarrow} := \sum_{j \neq i} R_{j \rightarrow i}$; $\mathbf{R}_{\rightrightarrows}$ denotes the vector with i th element $R_{i \rightrightarrows}$.

At Date 1, banks' liquidity shocks are realized.¹⁷ We write $\sigma_i = 1$ if B_i is shocked and $\sigma_i = 0$ otherwise. There is no other risk, so the "state," which is realized at Date 1, is the profile $\{\sigma_i\}_i =: \boldsymbol{\sigma}$; the number of shocked banks is denoted by $S =: \sum \sigma_i$. (We impose no restrictions on the shock distribution.) If the shock exceeds B_i 's total pledgeable assets,

$$\theta y + R_{i \leftrightsquigarrow} < \ell \sigma_i, \quad (1)$$

it repays zero to other banks and gets payoff zero itself.¹⁸ The non-pledgeable part of its asset $(1 - \theta)y$ is destroyed. We say B_i is "liquidated." Otherwise, it continues to Date 2.

Our definition of liquidation assumes that B_i 's interbank liabilities, which are due at Date 2, do not impede its ability to meet the shock, which occurs at Date 1. That contrasts

(directed paths between all pairs of banks), that of Lemma 7. And nothing in our efficiency benchmark relies on it: The planner in Definition 7 is free to choose any transfers, so a network that attains its solution—as the exponential network does—is best among all networks, zero-net or not.

¹⁶For example, in 2021, Barclays PLC's net interbank position was about an eighth of its gross, Lloyd's about a fourteenth, and HSBC's about a fifth. Specifically, their loans to and from other banks were, respectively, about 13.9 and 16.4, 7.0 and 7.6, and 83.1 and 101.1 billion GBP; see [home.barclays/content/dam/home-barclays/documents/investor-relations/reports-and-events/annual-reports/2021/Barclays-Bank-PLC-2021-AR.pdf](https://www.barclays.com/content/dam/home-barclays/documents/investor-relations/reports-and-events/annual-reports/2021/Barclays-Bank-PLC-2021-AR.pdf), p. 207, [lloyds-bankinggroup.com/assets/pdfs/investors/annual-report/2021/2021-lbg-annual-report.pdf](https://www.lloydsbankinggroup.com/assets/pdfs/investors/annual-report/2021/2021-lbg-annual-report.pdf), pp. 207–208, and [hsbc.com/investors/results-and-announcements/annual-report](https://www.hsbc.com/investors/results-and-announcements/annual-report), p. 310.

¹⁷Given the two-date structure, all liquidity shocks arrive at Date 1. What matters is not that they arrive at the same time, but rather while gross positions are still outstanding. That seems reasonable to the extent that interbank debts are long term. In contrast, dilution does nothing for shocks that arrive after the debts they would dilute have matured; see the discussion of random-time liquidity shocks in Donaldson and Micheler (2018).

¹⁸The assumption that interbank repayments are zero follows AOT, in which liquidity shocks—their "outside obligations"—are senior to interbank debts. The opposite assumption makes the repayment in equation (5) discontinuous and can lead to multiple payment equilibria (see Definition 1 below). It does not, however, alter our central insights on the benefits of high indebtedness and connectedness with long-term debt; see Donaldson and Piacentino (2018).

with the networks literature (see AOT and Section 3), but is in keeping with papers in corporate finance (the assumptions are close to Donaldson, Gromb, and Piacentino’s (2025)). It captures the idea that B_i can issue new liabilities of high priority, paid ahead of existing interbank liabilities at Date 2, which are thereby diluted, in that their claims on assets are now subordinated to new creditors’.¹⁹ These new liabilities could represent repos, which have “super-senior” claims on assets in bankruptcy.^{20,21} As a result, banks can borrow against all their pledgeable assets—the pledgeable part θy of their long-term assets and their interbank claims $R_{i\pm}$.²²

At Date 2, a bank that is not liquidated continues to produce y . As $(1 - \theta)y$ is not pledgeable, its total pledgeable assets are therefore $\theta y - \ell\sigma_i + R_{i\pm}$.²³ If they exceed its debt

¹⁹Note that dilution, as used here, need not decrease the value of existing liabilities. Although existing liabilities have a smaller slice of the asset pie, these assets might be more valuable if dilution increases the size of the pie (e.g., by helping to meet a liquidity shock).

²⁰As such, the model could reflect how banks suffering liquidity shocks use super-senior (repo) financing to relax their borrowing constraints, something LTCM, Bear Stearns, and Lehman Brothers all did (or tried to do). (See, e.g., Jorion (2000, pp. 282–284) on LTCM, Rose, Bergstresser, and Lane (2009, pp. 11–13) on Bear, and Valukas (2010, pp. 3 and 9–10) on Lehman.)

²¹Thus, there are two priority classes of debt in our model: new repo-type debt paid first and interbank debts paid next pro rata. This is a good approximation of reality, in which there are two main priority classes: secured debt paid first and unsecured paid next pro rata (see, e.g., Schwartz (1989)). (AOT also features two priority classes, but the senior debt is in place at inception.) What matters is that the new claim is senior, not that it is debt: A shocked bank can raise cash only against a claim that its existing counterparties cannot absorb first.

²²We interpret $(1 - \theta)y$ as cash that insiders can freely divert at Date 2—a pure private benefit, consumed by the bank—so creditors can reach at most θy in any state, in or out of default. In some contexts, this is a catch-all for richer agency frictions, such as asymmetric information as DeMarzo and Fishman (2007) show formally. However, the interpretation can matter for welfare. For example, if the private benefit were to stem from socially wasteful behavior, liquidation need not be inefficient, something we show in Section 8 below.

²³Two things are implicit behind this expression. As banks consume only at Date 2 and have no outside technologies, there is no use for funds raised at Date 1 except to meet liquidity shocks. In principle, we could allow them to borrow, hold cash, and repay. That would net out, so we abstract from it for notational simplicity.

You could imagine allowing for consumption at Date 1, which could capture, e.g., dividend payouts. We assume that is not allowed because the law places tight limits on payouts by distressed firms (e.g., unlawful dividend rules, fraudulent transfer law, and bankruptcy clawback provisions). That is not inconsistent with distressed firms issuing high priority debt, which is less restricted in practice (e.g., DIP financing, priming debt, or liability-management transactions). Moreover, bank debt in particular carries few covenants (Goyal (2005)) and negative pledge clauses, where present, are deemed ineffective (Bjerre (1999); Donaldson, Gromb, and Piacentino (2025)).

Our analysis below suggests a possible efficiency rationale for this asymmetry between limits on payouts and dilution, namely that coinsurance relies on banks being able to dilute existing creditors to raise liquidity, but not simply to expropriate value, which would lead the coinsurance mechanism to unravel.

$F_{i\Rightarrow}$, it repays and keeps the remainder of its total assets. Otherwise, i.e. if

$$\theta y - \ell\sigma_i + R_{i\Leftarrow} < F_{i\Rightarrow}, \quad (2)$$

it defaults, its creditors get its pledgeable assets, and it keeps its non-pledgeable assets. In summary, if not liquidated, B_i 's creditors get $\min\{\theta y - \ell\sigma_i + R_{i\Leftarrow}, F_{i\Rightarrow}\}$ and B_i gets $\max\{(1 - \theta)y, y - \ell\sigma_i + R_{i\Leftarrow} - F_{i\Rightarrow}\}$.

We assume that in default a bank's total pledgeable assets go to its counterparties with its total interbank repayments being divided in pro rata shares:

$$R_{i\rightarrow j} = \hat{F}_{i\rightarrow j} R_{i\Rightarrow}, \quad (3)$$

where $\hat{F}_{i\rightarrow j} := F_{i\rightarrow j}/F_{i\Rightarrow}$ if $F_{i\Rightarrow} \neq 0$ and $\hat{F}_{i\rightarrow j} := 0$ otherwise. The pro rata rule follows the literature (Eisenberg and Noe (2001)) and reflects bankruptcy law and practice.²⁴ (The identity also holds out of default, as $F_{i\rightarrow j} = \hat{F}_{i\rightarrow j} F_{i\Rightarrow}$.)

Whereas liquidation entails a deadweight loss (the only one in the model), default is just a transfer from creditors to debtors (except in extensions; see Section 8.1 and Section 8.2).

To define the equilibrium, we also require that markets clear: The repayments B_i receives from other banks coincide with the repayments other banks make to it:

$$R_{i\Leftarrow} = \sum_{j \neq i} R_{j \rightarrow i}. \quad (4)$$

Definition 1 (Payment equilibrium). *A payment equilibrium is a repayment vector $\{R_{i\rightarrow j}\}_{i \neq j}$ for each state σ such that the repayments*

²⁴Csóka and Herings (2021) provides an axiomatic foundation for the pro rata assumption.

(i) are sequentially rational, or, combining the above,

$$R_{i\Rightarrow} = \begin{cases} 0 & \text{if } \theta y - \ell \sigma_i + R_{i\Leftarrow} \leq 0, \\ \theta y - \ell \sigma_i + R_{i\Leftarrow} & \text{if } \theta y - \ell \sigma_i + R_{i\Leftarrow} \in (0, F_{i\Rightarrow}], \\ F_{i\Rightarrow} & \text{otherwise} \end{cases} \quad (5)$$

(ii) are paid pro rata (equation (3)), and

(iii) clear the market (equation (4)).

It is convenient to write the sequential rationality condition as

$$R_{i\Rightarrow} = \max \left\{ 0, \min \left\{ \theta y - \ell \sigma_i + R_{i\Leftarrow}, F_{i\Rightarrow} \right\} \right\}. \quad (6)$$

That can be combined with the other equilibrium conditions to write a vector fixed point equation:

$$\mathbf{R}_{\Rightarrow} = \left[\min \left\{ \mathbf{F}_{\Rightarrow}, \theta y \mathbf{1} - \ell \boldsymbol{\sigma} + \hat{\mathbf{F}}^{\top} \mathbf{R}_{\Rightarrow} \right\} \right]^+, \quad (7)$$

a solution of which is often called the “clearing vector.” ($\mathbf{1}$ denotes the vector of N ones: $(1, \dots, 1) \in \mathbb{R}^N$.)

As liquidation is the only inefficiency in the model, we adopt the following notion of efficiency.

Definition 2 (Efficiency). *One network is (weakly) more efficient than another if (weakly) fewer banks are liquidated in equilibrium for every state $\boldsymbol{\sigma}$.*

This is a strengthening of AOT’s notions of “stability” (fewer liquidations on average for a given number of shocks) and “resilience” (fewer liquidations in the worst case scenario) in that if one network is more efficient than another it is more stable and more resilient too. The efficiency ranking is not a total order on the set of networks, but we derive strong

enough results that it suffices for our purposes (except in the extension with heterogeneous banks in Section 7, where we modify it; see Definition 12).

For several of our results, it is useful to define the following network structures:

Definition 3 (Network typology). *A network $\mathbf{F}$ is regular if $F_{i\rightarrow} = F_{j\rightarrow}$ for $i \neq j$, symmetric if $F_{i\rightarrow j} = F_{j\rightarrow i}$ for $i \neq j$, complete if $F_{i\rightarrow j} > 0$ for $i \neq j$, uniform complete if it is complete and $F_{i\rightarrow j}$ is constant, star if $F_{i\rightarrow j} = F_{j\rightarrow i}$ is constant for some i and all j and $F_{j\rightarrow k} = 0$ for all $j, k \neq i$, and ring if $F_{i\rightarrow j} = 0$ unless $j = i + 1 \pmod{N}$.*

In words, in a regular network, each bank has the same total liabilities; in a symmetric network, each pair of banks has zero net positions; in a complete network, every pair of distinct banks is directly connected (every ordered pair has a positive exposure); in a uniform complete network, every pair of distinct banks is directly connected by a liability of the same size; in a star network, one bank—the “core”—owes and is owed the same amount by every other, and the others owe nothing to each other; in a ring network, each bank owes only one other bank and is owed by one other bank in a circle.

It is also useful to define several properties of networks, capturing how closely banks are connected to one another.

Definition 4 (Delta connectedness). *A network $\mathbf{F}$ is δ -connected if there is a non-empty proper subset of banks $\mathcal{B}$ such that $\hat{F}_{i\rightarrow j} \leq \delta$ and $\hat{F}_{j\rightarrow i} \leq \delta$ for all $i \in \mathcal{B}$ and $j \in \mathcal{B}^c$.*

It is connected if it is not δ -connected for $\delta = 0$.

In words, δ measures the strength of the weakest link between two parts of the network. Small δ means the network can be partitioned into two groups connected only by weak liabilities. At $\delta = 0$ the two groups have no liabilities between them at all, so a network that is *not* 0-connected is connected in the usual sense of graph theory: There is an undirected path between every pair of banks. Without loss, we focus on connected networks for the following definitions and results unless otherwise stated.

Definition 5 (Harmonic distance). *For a regular network $\mathbf{F}$, the harmonic distance from B_i to B_j is the solution to $d_{i \rightarrow j} := 1 + \sum_{k \neq i} d_{i \rightarrow k} \hat{F}_{k \rightarrow j}$ for $i \neq j$ and $d_{i \rightarrow i} = 0$.*

In words, $d_{i \rightarrow j}$ is the liability-weighted distance from i to j , which captures how easily liquidity (or the lack thereof) can flow from B_i to B_j : It is small when payments travel in few steps over links that carry a large share of each bank's liabilities. What matters is the share, not the count: A bank connected to many others by thin links can be “far” from all of them.

Definition 6 (Bottleneck parameter). *For a regular network $\mathbf{F}$, the bottleneck parameter is*

$$\beta = \min_{\emptyset \neq \mathcal{B} \subsetneq \{1, \dots, N\}} \sum_{i \in \mathcal{B}} \sum_{j \in \mathcal{B}^c} \frac{\hat{F}_{i \rightarrow j}}{|\mathcal{B}| |\mathcal{B}^c|}. \quad (8)$$

In words, β is the narrowest pipe out of any group of banks: A network has low β if one of its components has relatively low liabilities to the rest of it, so that little can drain out of that component. It is similar to δ above, but directional (in that it is silent about the liabilities from the rest to the component).

3 Benchmark Without Dilution

Here we consider the network in which the interbank liabilities $F_{i \rightarrow j}$ are due at Date 1 instead of Date 2. This benchmark helps us both to compare our model to the literature and to contrast our results to those therein.

Now interbank liabilities, being due immediately, cannot be diluted with new debt at Date 1. Thus B_i is liquidated if its pledgeable assets are insufficient to cover not only its liquidity needs $\ell \sigma_i$ but also its interbank liabilities $F_{i \rightrightarrows}$, or

$$\theta y + R_{i \Leftarrow} < \ell \sigma_i + F_{i \rightrightarrows}. \quad (9)$$

This condition for liquidation coincides with that for default. In that respect, the benchmark

contrasts with the baseline, in which the conditions are different. Nonetheless, the equations for the clearing vector are the same in both versions, as the liquidation value at Date 1 coincides with the pledgeable value at Date 2: Banks that would default at Date 2 in the baseline model make the same repayment when they are liquidated at Date 1 here. Thus we do not need to adjust the equilibrium definition here. Only efficiency changes.

Without the distinction between liquidation and default, this benchmark coincides with AOT.

Lemma 1 (Isomorphism between benchmark and AOT). *There is an isomorphism between payment equilibria in our benchmark and those in AOT in the case in which long-term assets are fully destroyed by default and cash holdings are zero.²⁵ The payment equilibria in the two models have the same liquidation outcomes; hence, they have generically the same efficiency ranking of networks.²⁶*

(The isomorphism between our benchmark and AOT, while mechanical, was unexpected to us, as the models seemed different *prima facie*. Our model seemed to be about liquidity, theirs about solvency. We now see both as about both: In both, a shock decreases total asset value (solvency) and might not be met only because long-term assets are not fully pledgeable (liquidity).)

We now restate, and sometimes strengthen, several of AOT’s results, focusing on those that contrast with our results on the long-term debt network.

Lemma 2 (Netting in benchmark). *For any network $\mathbf{F}$ and any $\alpha > 1$, $\alpha\mathbf{F}$ is generically less efficient than $\mathbf{F}$.*

This generalizes AOT’s Proposition 3 (p. 574) in two ways: To an arbitrary number of shocked banks (they prove it for one) and beyond regular networks. It says that less debt is

²⁵This is the case under which AOT derive most of their results.

²⁶Here and hereinafter, “generically” means everywhere except on the “knife-edge” set of parameters on which the equilibrium is not unique; see footnote 28 below for the specifics. The efficiency ranking in the lemma holds there too under any selection rule applied to both models.

a good thing. Indeed, it would be better to have none whatsoever ($\alpha = 0$). Intuitively, when one bank defaults on its liability to another, the other finds it harder to pay its liability to yet another. So distress propagates from shocked banks to otherwise healthy ones, especially when debts are high ($\alpha > 1$).

We now turn to network connectedness.

Lemma 3 (Delta connectedness in benchmark). *Suppose ℓ is sufficiently large and exactly one bank is shocked ($|\sigma| = 1$).*

- (i) *A ring network with $F > (N - 1)\theta y$ is the least efficient among all networks.*
- (ii) *Any δ -connected network with $N\delta \max_k F_{k\rightleftharpoons} < \theta y$ is strictly more efficient than the ring.*

This strengthens some of the statements in AOT’s Proposition 6 (p. 577–578) by adapting them to our notion of efficiency (Definition 2) and dropping regularity. It says that less connectedness is a good thing. The ring network, in which every bank has a large exposure to another, is the worst. It is better to weaken the exposures in the sense of lowering delta connectedness. Intuitively, δ captures how much risk can transmit between two components, so, unlike in the ring network, risk cannot spillover from one to another when δ is low.

We now show that there is a “default radius” around a shocked bank.

Lemma 4 (Default radius in benchmark). *Let $\mathbf{F}$ be a regular network with $F_{i\rightleftharpoons} \equiv F$ and suppose that exactly one bank, say B_j , is shocked and does not meet its liquidity shocks ($R_{j\rightleftharpoons} = 0$). Define $d^{ST} := \frac{F}{\theta y}$.*

- (i) *If $d_{j\rightarrow i} < d^{ST}$, then B_i is liquidated.*
- (ii) *If all banks are liquidated, then $d_{j\rightarrow i} < d^{ST}$ for all i .*

This is AOT’s Proposition 8 (p. 579). It says that the harmonic distance d is, in a sense, the right measure of one bank’s exposure to another, in that it captures exactly whether a

shocked bank’s distress will transmit to an otherwise healthy bank through the network. It defines a radius around a shocked bank within which all banks are liquidated.

AOT link the harmonic distance d to the bottleneck parameter β using Markov chains. They show, roughly, that $d_{i \rightarrow j}$ is the mean hitting time of a Markov chain from state i to j and that the bottleneck parameter is closely related to the “conductance” of a graph, which measures how hard it is for a Markov chain on a graph to leave a set of nodes. Hence the next result:

Lemma 5 (Bottleneck connectedness in benchmark). *Suppose the conditions of Lemma 4 are met and that, additionally, the network $\mathbf{F}$ is symmetric. Define $\beta^{ST} := 4\sqrt{\frac{\theta_y}{NF}}$ and $\beta_{ST} := \min \left\{ \frac{\theta_y}{2NF}, 1 \right\}$.*

(i) *If $\beta > \beta^{ST}$, then all banks are liquidated.*

(ii) *If $\beta < \beta_{ST}$, then at least one bank is not liquidated.*

This is AOT’s Corollary 2 (p. 581). It captures the idea that if all banks are closely connected then risk is so easily transmitted to other banks that a shock at one can lead all to fail, whereas if they are not, it cannot. Specifically, if at least two components are not closely connected, so β is small, risk in one of them cannot spread to the other.

4 Properties of Dilutable Debt Networks

We now turn to the properties of networks in our baseline model, in which interbank debt is dilutable. For each result in the non-dilutable benchmark, we prove a counterpart with the opposite sign: Whereas indebtedness and connectedness do harm when debt cannot be diluted, they do good when it can. The reason is that dilution changes what the network transmits—rather than propagating liquidity shortages from shocked banks, it lets liquidity flow from healthy banks to shocked ones.

We start with existence and uniqueness.

Proposition 1 (Existence and uniqueness). *For any network $\mathbf{F}$, a payment equilibrium exists and is generically unique.*

Existence, as ever, follows from Tarski’s fixed-point theorem. Uniqueness comes from payments being pinned down by resources, not expectations: Default is only a transfer (equation (2)), in which a bank hands its creditors all of its pledgeable resources, so its repayment moves at most one for one with the repayments it receives (equation (6))—a bank with a cushion absorbs part of any shortfall and a bank already repaying nothing passes on none of it. Any shortfall thus dissipates as it circulates through the network, no matter what banks expect of one another.²⁷ That said, there is a knife-edge case in which shocks exactly exhaust the total resources so repayments exactly cancel out, leading to multiple possible repayments supporting the same outcome.²⁸

We now turn to the role of debt levels.

Proposition 2 (Netting). *For any network $\mathbf{F}$, $\alpha\mathbf{F}$ is more efficient than $\mathbf{F}$ whenever $\alpha > 1$.*

This is the counterpart of Lemma 2. It says that *more* (dilutable) debt is a good thing here.²⁹ The reason is that high debts ($\alpha > 1$) allow shocked banks to weather shocks. To see why, suppose two banks, B_i and B_j , have perfectly offsetting debts, owing each other the

²⁷In related models, an equilibrium in which all banks in a cycle repay, and hence each has the resources to repay, can coexist with a default equilibrium with the opposite outcome. There, default is costly: A counterparty’s default inflicts a discrete loss on its creditors, so a shortfall compounds as it circulates, and pessimism around a cycle of claims validates itself (Rogers and Veraart (2013); Jackson and Pernoud (2024)); relatedly, in Roukny, Battiston, and Stiglitz (2018) banks’ default conditions depend on their counterparties’ defaults around a credit cycle. Liquidation destroys value in our model too, but only the non-pledgeable part $(1 - \theta)y$, which creditors could never have reached, so not even liquidation inflicts a discrete loss on creditors. (That relies on liquidity shocks being senior to interbank debts; see footnote 18.)

²⁸For a connected network, the knife-edge condition is $N\theta y = S\ell$. More generally, in a disconnected zero-net-position network, multiplicity can arise only if $|K|\theta y = S_K\ell$ for some connected component K where S_K is the number of shocked banks in K . To see where it comes from, suppose two banks, say B_i and B_j , owe each other F and have no outside liabilities. Knife-edge multiplicity can arise in the case in which they both default. Then, from equation (6), $R_{i \rightarrow j} = \theta y - \ell\sigma_i + R_{j \rightarrow i}$ and $R_{j \rightarrow i} = \theta y - \ell\sigma_j + R_{i \rightarrow j}$. Adding equations says any solution must satisfy $2\theta y = (\sigma_i + \sigma_j)\ell$, the two-bank version of the knife-edge condition. In this case, any pair of repayments such that the difference in payments equals the difference in liquidity shortfalls, $R_{i \rightarrow j} - R_{j \rightarrow i} = \theta y - \sigma_i\ell = \sigma_j\ell - \theta y$ is a solution. In particular, if $\sigma_i = \sigma_j = 1$ then the last condition holds for any $R_{i \rightarrow j} = R_{j \rightarrow i} \in [0, F]$: The payments cancel out, so it does not matter what they are.

²⁹Although, per the result, increasing debt cannot hurt if the relative debts stay the same—if $\alpha\mathbf{F}$ increases but $\mathbf{F}$ stays constant—it can if they change—i.e. if some debts increase and some do not—as illustrated by how making a network “more symmetric” can decrease efficiency (in Section 5).

same amount: $\alpha F_{i \rightarrow j} = \alpha F_{j \rightarrow i} = \alpha F$. Suppose B_i suffers a liquidity shock and B_j does not. In this case, B_i gets liquidity by pledging both its own assets and the claim it has on B_j to raise $\theta y + R_{j \rightarrow i}$. The larger B_i 's claim αF on B_j is, the higher its value $R_{j \rightarrow i}$ is, and the more liquidity it can raise.

True, increasing α gives B_i not only a larger claim to pledge on the left-hand side of its balance sheet, but also a larger liability to repay on the right. However, B_i , being shocked, is likely to default, whereas, not being shocked, B_j is not.³⁰ Therefore, the claim B_i has on B_j is worth more than the liability it has to it. So the apparently offsetting debts are a transfer of value from the not-shocked to the shocked bank, providing insurance for the shocked bank.

This transfer benefits whichever bank is shocked at the expense of whichever is not shocked. Liquidity gets transferred from the bank with liquidity to the bank that needs it: Zero-net long-term debt has positive net present value. Per the result, increasing debt increases overall efficiency.³¹

That is not so in the benchmark (and most other models), in which zero-net debts have zero NPV at most (Lemma 2). The reason is that short-term debt, being due right away, cannot be diluted: The claims on the left side of bank balance sheets increase debt capacity but the liabilities on the right decrease it; they cancel each other out at best.

We turn to network connectedness next.

Proposition 3 (Delta connectedness). *Suppose exactly one bank is not shocked ($|\mathbf{1} - \boldsymbol{\sigma}| = 1$).*

(i) *A ring network with $F > \theta y$ is the most efficient among all networks.*

³⁰Indeed, the shocked bank generally defaults, whereas the not-shocked one repays in full (cf. equation (2)): The shocked bank's default passes B_j a shortfall $\ell - \theta y$, which B_j 's own slack θy covers given $\ell < 2\theta y$ (the algebra is that of footnote 28 with one shock). Moreover, the mechanism is self-enforcing, in that, as banks cannot avoid being diluted, they are committed ex ante to transfers they would prefer not to make ex post. Leitner (2005) uncovers another self-enforcing mechanism to transfer liquidity in financial networks: Healthy banks commit to transfer liquidity to distressed ones by exposing themselves to their default through the network.

³¹Efficient "dilutable debt" also appears in Diamond (1993), Donaldson, Gromb, and Piacentino (2020, 2025), and Hart and Moore (1995).

(ii) If $\ell \leq 2\theta y$,³² for any δ , there is a δ -connected network that is strictly less efficient than a ring.

This is the counterpart of Lemma 3. It says that *more* connectedness is a good thing—connectedness in the sense of the bottleneck δ , which captures the strength of exposures between groups of banks, not their number. The ring network, in which every bank has a large exposure to another, is the best (given a single not-shocked bank). It is better not to weaken interbank exposures in the sense that making δ small could lead to a strictly worse outcome. Intuitively, δ captures how easily a shocked bank can raise liquidity using claims on healthy banks as collateral. When δ is low, that mechanism is less effective. The ring wins here because it sends the not-shocked bank’s liquidity to a single shocked bank rather than splitting it—the principle the exponential network generalizes to every state (Section 5).

We turn to the long-term-debt counterpart to short-term debt’s default radius around a shocked bank. It is a “salvation radius” around a healthy bank.

Proposition 4 (Salvation radius). *Let $\mathbf{F}$ be a regular network with $F_{i\rightleftharpoons} \equiv F$ and suppose that exactly one bank, say B_j , is not shocked and that it does not default. Define $d^{LT} := \frac{F}{\ell - \theta y}$.*

- (i) *If $d_{j \rightarrow i} < d^{LT}$, then B_i is not liquidated.*
- (ii) *If no bank is liquidated, then $d_{j \rightarrow i} \leq d^{LT}$ for all i .*

This is the counterpart of Lemma 4. It says that the harmonic distance captures not only how defaults transmit from shocked to healthy banks when debt cannot be diluted, but also how the option to dilute allows shocked banks to raise liquidity against claims on not-shocked banks. It defines a radius around a not-shocked bank within which no bank is liquidated.

AOT’s link between the harmonic distance and the bottleneck parameter also applies to our long-term debt network—it relies on only network structures, not equilibrium behavior.

Hence we have the next result:

³²This is the condition for there to be enough liquidity in total to save a shocked bank; otherwise no network does better than the empty network (cf. equation (10)).

Proposition 5 (Bottleneck connectedness). *Suppose the conditions of Proposition 4 are met and that, additionally, the network $\mathbf{F}$ is symmetric. Define $\beta^{LT} := 4\sqrt{\frac{\ell - \theta y}{NF}}$ and $\beta_{LT} := \min\left\{\frac{\ell - \theta y}{2NF}, 1\right\}$.*

(i) *If $\beta > \beta^{LT}$, then no bank is liquidated.*

(ii) *If $\beta < \beta_{LT}$, then at least one bank is liquidated.*

This is the counterpart of Lemma 5. It captures the idea that if all banks are closely connected, one bank's excess liquidity can flow through the system to save all banks and, conversely, if they are not, it cannot. Specifically, if at least two components are not closely connected (β is small), the banks in one can raise little liquidity by diluting their liabilities to banks in the other. What they can raise can be so limited that they end up unable to save themselves from liquidation.

5 Efficiency and Exponential Networks

Here we define constrained efficiency and construct a class of networks—the “exponential networks”—that implement it. We conclude the section with an example contrasting complete and exponential networks, which captures the main ideas.

Definition 7 (Planner's problem and constrained efficiency). *The planner's problem is to find a set of transfers $\{t_i\}_i$ for each σ , with t_i paid to each B_i , to minimize the number of liquidated banks $|\{i : \theta y + t_i - \ell \sigma_i < 0\}|$ subject to each bank's liquidity constraint $t_i \geq \min\{\ell \sigma_i - \theta y, 0\}$ and to liquidity being conserved $\sum t_i \leq 0$.*

A network is constrained efficient if the equilibrium is no less efficient than the planner's solution.

In words, the planner wants to minimize the number of liquidated banks by transferring liquidity within the system. It must respect the aggregate resource constraint and the limited pledgeability friction—it cannot raise more from any one bank than the net liquidity it has

(that is no more than θy from a not-shocked bank and zero from a shocked one, per the constraint $t_i \geq \min\{\ell\sigma_i - \theta y, 0\}$, given $\ell > \theta y$). But it has no other constraints. (The problem does not even depend on the network $\mathbf{F}$ as the planner's transfers can replicate—or undo—any interbank payments.) The solution, characterized next, is triage: Tax every healthy bank to the hilt, fully rescue as many shocked banks as that budget allows, and abandon the rest, as spreading liquidity thinly saves no one.

Lemma 6 (Constrained efficiency). *A network is constrained efficient if the number of liquidated banks is*

$$L^* := \max \left\{ 0, \left\lceil \frac{S\ell - N\theta y}{\ell - \theta y} \right\rceil \right\} \quad (10)$$

for each state σ where S banks are shocked.

It turns out that the social planner should generally raise as much liquidity as possible from each not-shocked bank, levying the tax $-t_i = \theta y$ if $\sigma_i = 0$, and transfer shocked banks either just enough liquidity to survive or none at all, i.e., either $t_i = \ell - \theta y$ or $t_i = 0$ if $\sigma_i = 1$. Using S and L to denote the numbers of shocked and liquidated banks, the planner's budget constraint says that the total subsidy—the transfer $\ell - \theta y$ to each of the $S - L$ shocked banks that is not liquidated—must be less than the total tax—the transfer θy from each of the $N - S$ banks that is not shocked:

$$(S - L)(\ell - \theta y) \leq (N - S)\theta y. \quad (11)$$

Solving for the smallest non-negative integer L that satisfies the above gives the result.

That argument points to two key properties of the planner's solution, both of which help avoid “wasting liquidity”:

- (i) It extracts the maximum tax from not-shocked banks, since they are not liquidated anyway.
- (ii) It gives nothing to liquidated banks, since, analogously, they are liquidated anyway.

We aim to construct a network with both properties (Proposition 6 below). We now build up to it in steps, showing how to achieve one and then the other, starting with the first:

Lemma 7 (High debt mutualizes assets). *Let $\mathbf{F}$ be a connected network. If α is sufficiently large, then in the equilibrium of $\alpha\mathbf{F}$ either (i) all not-shocked banks make the maximum net payment, $R_{i\rightarrow} - R_{i\leftarrow} = \theta y$, or (ii) no bank is liquidated.*

This says that if debts are sufficiently high and liquidity is sufficiently scarce (in the sense that at least one bank is liquidated), then each not-shocked bank provides the maximum amount of liquidity. Intuitively, to increase interbank debts is to make each bank’s assets a larger fraction of others’ balance sheets—it “mutualizes” the banking system, making each bank more like the whole system.³³ As a result, when liquidity is scarce overall, no surviving bank retains excess liquidity. In contrast, if liquidity is not scarce then no bank is liquidated if debt levels are sufficiently high:

Corollary 1 (First best with high debt for small shocks). *Let $\mathbf{F}$ be a connected network and suppose that $N\theta y > S\ell$ (i.e. $L^* = 0$). If α is sufficiently large, then no bank is liquidated in the equilibrium of $\alpha\mathbf{F}$.*

The second property—that liquidated banks are transferred nothing—points to how the planner’s allocation is necessarily discriminatory: It prioritizes some shocked banks over others. (If in contrast, it allocated the excess liquidity equally among all shocked banks it ends up saving none of them unless it can save them all. As we illustrate in an example below, that symmetry makes the complete network “robust yet fragile,” per AOT’s result.) That suggests that whenever liquidity is scarce (S is large) a network must be asymmetric to be (constrained) efficient. The following definitions characterize ways in which a network can be asymmetric.

³³This role of default in facilitating a socially efficient transfer of liquidity contrasts with the literature, in which it typically constitutes a social cost, facilitating rent extraction at best (see Farboodi (2023) and Perotti and Spier (1993)).

Definition 8 (Assortativity). *A network $\mathbf{F}$ is assortative if there is a permutation π on $\{1, \dots, N\}$ such that*

$$F_{i \rightarrow \pi(k)} \geq F_{i \rightarrow \pi(l)} \quad (12)$$

for all $k < l$ and all $i \notin \{\pi(k), \pi(l)\}$.

In words, a network is assortative if whenever one bank owes more to B_k than to B_l , every other bank owes weakly more to B_k than to B_l . Assortativity thus allows us to rank banks by the size of the liabilities they are owed: The network has a single weak hierarchy on which all banks agree, with ties allowed, rather than each bank owing most to its own neighbors as in, e.g., the ring. The next definition quantifies how steep that hierarchy is.

Definition 9 (s -dominance). *For a network $\mathbf{F}$, B_i 's liabilities are s -dominated for $s \in (0, 1)$, if there is a permutation π_i on $\{1, \dots, N\}$ with $\pi_i(i) = i$, such that*

$$F_{i \rightarrow \pi_i(j+k)} \leq s^k F_{i \rightarrow \pi_i(j)} \quad (13)$$

for all j and $k \geq 0$ such that $j \neq i$ and $j + k \neq i$.

In words, for $s < 1$, B_i 's liabilities to others decay rapidly—its second largest liability is only at most a fraction s of its largest, and so on: The pecking order is steep, with each bank's payments concentrated at the top of its ranking rather than spread down it. (The permutation π_i in the definition just ranks the debts by size.)

Together the definitions above define what we call exponential networks:

Definition 10 (Exponential networks). *A network is an exponential network (with base s) if it is connected, it is assortative, and B_i 's debts are s -dominated for all i .*

In words, every bank's debt to B_2 is only at most a fraction s of its debt to B_1 and so on—the permutation π_i in Definition 9 can be chosen to rank each bank's creditors the same way (assumed w.l.o.g. to be the same as their index ordering per Definition 8), creating a pecking

order: Liquidity flows to the top-ranked banks first, rather than being divided evenly and wasted, as we show below. The exponential network generates an approximately exponential distribution of bank asset size, $y + F_{i\Leftarrow}$.³⁴

The next two results characterize the payments made to liquidated banks in an exponential network.

Lemma 8 (Controlling relative payments to liquidated banks). *Suppose $\mathbf{F}$ is an exponential network with base s and B_{i^*} is liquidated. For every lower-ranked bank B_j , $j > i^*$,*

$$R_{j\Leftarrow} \leq s^{j-i^*} R_{i^*\Leftarrow}. \quad (14)$$

This says that when debt levels in the network are exponentially controlled (given s -dominance), so are the repayments to liquidated banks in equilibrium.

The previous result says that s controls the relative payments among liquidated banks. The next says that it controls the total payment to all of them.

Lemma 9 (Controlling total payments to liquidated banks). *Suppose $\mathbf{F}$ is an exponential network with base s . If at least one bank is liquidated in equilibrium, then*

$$\sum_{i \in \mathcal{L}} R_{i\Leftarrow} < \frac{\ell - \theta y}{1 - s}, \quad (15)$$

where $\mathcal{L}$ denotes the set of liquidated banks.

This says that the total transfer to liquidated banks can be made arbitrarily close to a single shocked bank's liquidity shortfall, $\ell - \theta y$, by making s sufficiently small. Intuitively, for small s , the liquidity wasted by transferring it to banks that end up being liquidated anyway would barely have been enough to save even a single one of them (per property (ii) above).

To sum up, if debts are large then no liquidity is wasted on banks that would not have been liquidated anyway (Lemma 7) and if the network is exponential then little is wasted

³⁴Another paper in which intermediation networks give rise to an endogenous bank size distribution is Farboodi, Jarosch, and Shimer (2017).

on those that would have been (Lemma 9). The next result builds on these findings to show how to achieve constrained efficiency.

Proposition 6 (Efficiency of exponential networks). *Define*

$$s^* := 1 - \frac{\ell - \theta y}{N\theta y - S\ell + (1 + L^*)(\ell - \theta y)}. \quad (16)$$

Let $\mathbf{F}$ be an exponential network with base $s \leq s^$. For α sufficiently large, $\alpha\mathbf{F}$ is constrained efficient as long as $\frac{S\ell - N\theta y}{\ell - \theta y}$ is not an integer.*

In words, an exponential network with a rapidly decaying distribution of debts is optimal in all but a knife-edge case (i.e. all but the case in which $\frac{S\ell - N\theta y}{\ell - \theta y}$ is an integer).

This result has a limitation when achieving constrained efficiency requires almost all of the liquidity available, i.e. when the slack in inequality (11) becomes small for $L = L^*$ ($((N - S)\theta y - (S - L^*)(\ell - \theta y)) \rightarrow 0$). In this case, s^* becomes small so the largest bank in the exponential network is many times larger than the second largest one, and so on. Moreover, in the limit when it equals zero, the network does not achieve constrained efficiency—that is the integer case from the proposition. But the next result suggests this limitation is not too worrisome if we accept a weaker notion of efficiency:

Proposition 7 (Approximate efficiency of exponential networks). *Let $\mathbf{F}$ be an exponential network with base $s \leq 1/2$. For α sufficiently large, $\alpha\mathbf{F}$ is “almost constrained efficient” in that at most $L^* + 1$ banks are liquidated.*

Finally, we point out that even in the knife-edge case in which the exponential network does not achieve constrained efficiency, no other fully connected network does either.

Lemma 10 (Inefficiency of other networks). *Suppose $\frac{S\ell - N\theta y}{\ell - \theta y}$ is an integer, $S < N$, and $S\ell > N\theta y$ and that $\mathbf{F}$ is fully connected in that $F_{i \rightarrow j} > 0$ for all $i \neq j$. The equilibrium is not constrained efficient.*

This result suggests that the exponential network is the “best” no matter the parameters.³⁵

Efficiency example. Finally, we consider an example to illustrate (i) how a complete network leads to inefficient liquidation by allocating liquidity to banks that end up being liquidated in equilibrium and (ii) how an exponential network solves the problem.

The illustration requires that shocks are large enough that at least one bank is liquidated in the constrained-efficient outcome (otherwise a complete network can save all banks). Hence we consider three banks, two of which are shocked: There are $N = 3$ banks with assets $y = 2$, a fraction $\theta = 1/2$ of which is pledgeable. Exactly two banks suffer shocks, $S \equiv \sum \sigma_i = 2$, of size ℓ . We assume that $\ell = 8/5$ so that in the constrained-efficient outcome exactly one bank is liquidated in each state: $\ell < 3\theta y < 2\ell$ (i.e. $8/5 < 3 < 16/5$). Observe that covering either shocked bank’s liquidity shortfall requires at least $3/5$ of the other’s surplus: $\ell - \theta y = 3\theta y/5$.

We start with the complete network as benchmark and show that both shocked banks are always liquidated. Then we illustrate how the exponential network saves one of them, achieving the constrained-efficient outcome.

Complete network benchmark. Suppose each bank has total liabilities $F_{i\rightarrow} \equiv F$ to others ($F/2$ to each of the other two). For each state σ , equation (5) gives the system the clearing vector $\mathbf{R}_{\rightarrow} = [R_{i\rightarrow}]_i$ must solve (the full matrix of equilibrium repayments is then given by the pro rata shares $R_{i\rightarrow j} = \hat{F}_{i\rightarrow j} R_{i\rightarrow}$ by equation (3)). When the shocked banks are B_1 and B_2 , the system is:

$$\begin{cases} R_{1\rightarrow} = \max \left\{ 0, \min \left\{ -\frac{3}{5} + \frac{1}{2}R_{2\rightarrow} + \frac{1}{2}R_{3\rightarrow}, F \right\} \right\}, \\ R_{2\rightarrow} = \max \left\{ 0, \min \left\{ -\frac{3}{5} + \frac{1}{2}R_{1\rightarrow} + \frac{1}{2}R_{3\rightarrow}, F \right\} \right\}, \\ R_{3\rightarrow} = \max \left\{ 0, \min \left\{ 1 + \frac{1}{2}R_{1\rightarrow} + \frac{1}{2}R_{2\rightarrow}, F \right\} \right\}. \end{cases} \quad (17)$$

³⁵As the proof of Lemma 10 implies, a network that is not fully connected could do better for some realizations of σ but not for all.

The first two lines of the system illustrate the problem with the complete network. The repayments from the bank with excess liquidity, B_3 , are allocated equally between the two banks with a liquidity shortfall, B_1 and B_2 —each gets $\frac{1}{2}R_{3\Rightarrow}$ (plus a symmetric transfer from the other shocked bank, $\frac{1}{2}R_{2\Rightarrow}$ or $\frac{1}{2}R_{1\Rightarrow}$). But each bank needs more than that to survive. Allocating scarce resources equally means that no one has enough.

As the system is symmetric, the problem is analogous when the other pairs of banks are shocked. Solving it gives $R_{i\Rightarrow} = 0$ if $\sigma_i = 1$ and $R_{i\Rightarrow} = \min\{F, 1\}$ otherwise, implying that both shocked banks are always liquidated (equation (5)).

Exponential network. Now we turn to an exponential network with base $s = 1/2$:

$$\mathbf{F} = \begin{bmatrix} 0 & 2 & 1 \\ 2 & 0 & \frac{1}{2} \\ 1 & \frac{1}{2} & 0 \end{bmatrix}. \quad (18)$$

Note that the off-diagonal entries in each row and column are decreasing (assortativity per Definition 8) and that each is at most a fraction $1/2$ of the previous (s -dominance per Definition 9). We can compute each bank's total liabilities $F_{i\Rightarrow}$ (the row sums of $\mathbf{F}$) and the fraction of its payments it makes to each other bank ($\mathbf{F}$ normalized by $F_{i\Rightarrow}$):

$$\mathbf{F}_{\Rightarrow} = \begin{bmatrix} 3 \\ \frac{5}{2} \\ \frac{3}{2} \end{bmatrix} \quad \& \quad \hat{\mathbf{F}} = \begin{bmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{4}{5} & 0 & \frac{1}{5} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{bmatrix}. \quad (19)$$

It turns out that no matter what pair of banks is shocked, only one is liquidated. To see why, consider the case in which B_1 and B_2 are shocked. In that case, the clearing vector solves (substituting from equation (19) into equation (5)):

$$\begin{cases} R_{1\Rightarrow} = \max \left\{ 0, \min \left\{ -\frac{3}{5} + \frac{4}{5}R_{2\Rightarrow} + \frac{2}{3}R_{3\Rightarrow}, 3 \right\} \right\}, \\ R_{2\Rightarrow} = \max \left\{ 0, \min \left\{ -\frac{3}{5} + \frac{2}{3}R_{1\Rightarrow} + \frac{1}{3}R_{3\Rightarrow}, \frac{5}{2} \right\} \right\}, \\ R_{3\Rightarrow} = \max \left\{ 0, \min \left\{ 1 + \frac{1}{3}R_{1\Rightarrow} + \frac{1}{5}R_{2\Rightarrow}, \frac{3}{2} \right\} \right\}. \end{cases} \quad (20)$$

The first two lines of the system illustrate how the exponential network allocates liquidity efficiently. The repayments from B_3 , the bank with excess liquidity, are allocated primarily to one of the two banks with the liquidity shortfall— B_1 gets $\frac{2}{3}$ of B_3 's total repayment, B_2 only $\frac{1}{3}$ of it. And that allows B_1 to survive. B_2 is liquidated. But that is (constrained) efficient as there are not enough resources to save them both anyway.

Solving the system gives the clearing vector $\mathbf{R}_{\Rightarrow} = (\frac{3}{35}, 0, \frac{36}{35})$, which, having only one zero entry, affirms that only B_2 is liquidated (equation (5)).

The other cases are analogous. When B_1 and B_3 are shocked the clearing vector is $\mathbf{R}_{\Rightarrow} = (\frac{3}{7}, \frac{9}{7}, 0)$, implying only B_2 is liquidated, and when B_2 and B_3 are shocked it is $\mathbf{R}_{\Rightarrow} = (\frac{39}{35}, \frac{1}{7}, 0)$, implying only B_1 is.

Ring network. Consider also the ring network, in which each B_i owes F to $B_{i+1} \pmod{3}$ and nothing to the other bank. It is also constrained efficient in this example: When B_1 and B_2 are shocked, the clearing vector is $\mathbf{R}_{\Rightarrow} = (\frac{2}{5}, 0, 1)$ for $F \geq 1$ —the not-shocked bank B_3 makes the maximum payment, all of it to B_1 , which survives—and the other states are analogous. But efficiency here is an artifact of the three-bank example, in which the shocked banks are necessarily adjacent. In Appendix B, we show that with four banks and a larger shock, the shocked banks can be separated by the not-shocked ones, in which case each not-shocked bank pays a different shocked bank and neither is saved.³⁶

³⁶Regularity and constrained efficiency generally push in opposite directions: One requires symmetry in banks' total interbank debts, the other asymmetry in which shocked banks are liquidated. The comparison of the complete network (regular) and exponential network (not) captures that. The ring with $N = 3$ is a caveat, as it can be both regular and constrained efficient. But, as Appendix B shows, this need not remain true once $N \geq 4$.

6 Pairwise Stability

We now explore what networks can arise through decentralized contracting. We start by defining a notion that captures that: Do banks have the incentive to deviate from their links?

Definition 11 (Pairwise stability w.r.t. G). *Fix a distribution G of shocks σ . A network $\mathbf{F}$ is pairwise stable if no pair B_i and B_j of banks with a direct link ($F_{i \rightarrow j} > 0$ or $F_{j \rightarrow i} > 0$) can change their face values in a way that strictly increases both of their expected payoffs.*

In words, a network is pairwise stable if it does not admit bilateral deviations: No two linked banks can privately change their mutual debts and both come out strictly ahead. Netting out is always an admissible deviation, corresponding to reducing offsetting face values. As such, a network with gross positions being pairwise stable captures banks choosing not to net out.

The definition is weak in that deviations must be strict—ties are broken toward the status quo—and some types of conceivable deviations are not allowed—no new links can be formed. (Deviations need not be symmetric, so, unlike in the baseline, we do not impose zero-net positions; Proposition 1 still applies, as its proof does not use that assumption.) Nonetheless, our first result here is negative: The exponential network, despite being efficient, is not pairwise stable:

Proposition 8 (Exponential not pairwise stable). *Suppose $\ell < 2\theta y$, $1 \leq L^*$, B_{N-1} and B_N are linked, and G puts non-zero weight on exactly one of them being shocked (i.e. $\mathbb{P}[\sigma_i(1 - \sigma_j) = 1] > 0$ for $(i, j) \in \{(N-1, N), (N, N-1)\}$).*

Let $\mathbf{F}$ be an exponential network with $s \leq s^$ and α be sufficiently large. $\alpha\mathbf{F}$ is not pairwise stable w.r.t. G (where L^* and s^* are as in equations (10) and (16)).*

The proposition illustrates that constrained efficiency does not imply pairwise stability. The same priority ordering that maximizes liquidity insurance creates profitable bilateral

deviations. In particular, B_{N-1} and B_N , who, by construction, always receive liquidity last in the exponential network, can trap some liquidity between them by increasing the gross debt between them. That suggests that the networks that arise via decentralized contracting are likely to be inefficient.

Hence we turn to networks that resemble those observed in the real world, which are largely found to be core-periphery networks. We take two approaches to capturing the core-periphery networks, corresponding to different interpretations of the banks in our model, both defined in Definition 3. The first is the star network, which is the simplest core-periphery network. The second is the uniform complete network, in which all pairs of banks are connected by equal debts. This is not a core-periphery network in itself. It captures the core-periphery structure to the extent that the core is connected and the long-term assets y and liquidity shocks ℓ in our model capture claims and liabilities on the periphery.

Proposition 9 (Star pairwise stable). *For any G s.t. $1 < S < N$, $L^* < S$, $S\ell > (2N - S)\theta y$, and $(N - S)F \geq (\ell - \theta y)$, the star network is pairwise stable w.r.t. G .*

This suggests that our model can explain observed interbank network structures. The result is strong in that it holds for all shock distributions G . The proof exploits the high-debt region in which the core bank always defaults. Hence we prove an auxiliary result in which it does not, suggesting the conclusion is robust:

Proposition 9' (Star still pairwise stable). *Suppose $\mathbf{F}$ is a star network of $N \geq 3$ banks, $S = N - 1$, and*

$$\max \left\{ \frac{\theta y}{N - 1}, \frac{(2\theta - 1)y}{N - 2} \right\} < F = \ell - \theta y < \frac{\theta y}{N - 2}. \quad (21)$$

The star network is pairwise stable w.r.t. the uniform distribution of shocks (every σ s.t. $|\sigma| = N - 1$ is equally likely).

The proof of this version relies on banks being on the brink of liquidation: Increasing pairwise debts would harm them, because they would be liquidated when they are shocked. That sounds more knife-edge than we think it is. As our model is bang-bang—you are liquidated

or not—that is the analog of a first-order condition—you take on offsetting debts until the risk of liquidation from the liability outweighs the insurance benefit of the claim.

The complete network is likewise pairwise stable:

Proposition 10 (Uniform complete pairwise stable). *Suppose $N \geq 3$ and $\ell > N\theta y$. Then there exists $\bar{F}$ such that for every $F \geq \bar{F}$, the uniform complete network with common face value F is pairwise stable w.r.t. G for all G .*

The proof of this result, which requires all banks be highly indebted ($F \geq \bar{F}$), exploits that shocked banks are liquidated and not-shocked banks default, offering a possible explanation for why banks are indebted and crises are systemic: It deters deviations in networks.

Pairwise stability says that, per Definition 11, no pair of banks can do anything to improve their payoffs. The outcomes are nonetheless inefficient for banks collectively. In both cases, the networks are “too symmetric,” in that they often allocate liquidity equally to banks instead of prioritizing them, violating the “no wasting liquidity” principle (see Section 5). In the star network, only the core bank can be saved. Other shocked banks are always liquidated:

Proposition 11 (Star inefficient). *Suppose $N\theta y < S\ell$. Let $\mathbf{F}$ be a star network. All shocked banks are liquidated if either (i) the core bank is not shocked or (ii) it is shocked and $(N - S)F < (\ell - \theta y)$. It is not constrained efficient whenever $L^* < S$.*

The uniform complete network can be even worse. There, all shocked banks are liquidated whenever shocks are large:

Proposition 12 (Uniform complete inefficient). *Let $\mathbf{F}$ be the uniform complete network in Proposition 10. Whenever*

$$S\ell > N\theta y, \tag{22}$$

all shocked banks are liquidated.

This says that whenever the constrained efficient number of liquidations L^* is not zero, all shocked banks are liquidated. (That can be seen from substituting the condition (22) on the number of shocked banks into the definition of L^* in equation (10).) Even though, per Section 4, dilutable debt necessarily helps, the result still affirms the idea that observed financial networks are “robust yet fragile,” do well for small shocks but struggle to sustain large ones.

7 Heterogeneous Banks

So far, we assumed that all banks had the same assets in place and the same size liquidity shocks. Now we allow them to be heterogeneous, with y_i denoting B_i ’s assets and $\ell_i > \theta y_i$ the size of its potential liquidity shock. We denote the deadweight loss of liquidating B_i by $\Delta_i \geq 0$; in the baseline it is the non-pledgeable value destroyed, $\Delta_i = (1 - \theta)y_i$.

We start with a generalization of the planner’s problem to this environment:

Definition 12 (Generalized planner’s problem (PP)). *The (generalized) planner’s problem is to find a vector of transfers $\mathbf{t}$ for each $\boldsymbol{\sigma}$, with t_i paid to each B_i , to minimize the total deadweight loss of liquidation,*

$$\text{minimize } \sum_{i=1}^N \mathbb{1}_{\{\theta y_i + t_i < \ell_i \sigma_i\}} \Delta_i, \quad (23)$$

subject to (i) each bank’s liquidity constraint

$$t_i \geq \min\{\ell_i \sigma_i - \theta y_i, 0\} \quad (24)$$

and (ii) total liquidity being conserved

$$\sum_{i=1}^N t_i \leq 0. \quad (25)$$

Observe that if Δ_i is the same for all banks, then the objective is just to minimize the number of liquidations. If $\ell_i - \theta y_i$ is also the same, the constraints are too and the problem coincides with the planner's problem in Definition 7.

We next show that the problem of choosing transfers among banks is equivalent to that of choosing the set of banks to liquidate, with effectively the same objective and subject to the same constraint on aggregate liquidity:

Definition 13 (Knapsack problem (KP)). *Find a vector of binary variables $\mathbf{x} \in \{0, 1\}^N$ for each σ , with $x_i = 0$ if B_i is liquidated, to minimize the deadweight loss,*

$$\text{minimize } \sum_{i=1}^N (1 - x_i) \Delta_i, \quad (26)$$

subject to liquidity being conserved

$$\sum_{i=1}^N x_i \sigma_i (\ell_i - \theta y_i) \leq \sum_{i=1}^N (1 - \sigma_i) \theta y_i. \quad (27)$$

In computer science, this problem is called the “knapsack problem,” as it describes a problem of finding the most valuable set of objects to put in a knapsack subject to a constraint on their total weight, just as ours describes finding the most valuable set of banks to save subject to a constraint on the total liquidity required, an equivalence formalized in the next result:

Proposition 13 (The planner's problem is the knapsack problem). *PP in Definition 12 is equivalent to KP in Definition 13 in that (i) if $\hat{\mathbf{t}}$ solves PP then $\hat{\mathbf{x}}$ with $\hat{x}_i := \mathbb{1}_{\{\theta y_i + \hat{t}_i \geq \ell_i \sigma_i\}}$ solves KP and (ii) if $\check{\mathbf{x}}$ solves KP then $\check{\mathbf{t}}$ with $\check{t}_i := \sigma_i \check{x}_i (\ell_i - \theta y_i) - (1 - \sigma_i) \theta y_i$ solves PP.*

The knapsack problem is hard to solve (it is NP-hard, in the language of computer science). Hence algorithms have been developed that deliver approximate solutions quickly. One is the “greedy algorithm,” defined in our context as follows:

Definition 14 (Greedy algorithm). *Suppose banks are ranked by a permutation π^{-1} on $\{1, \dots, N\}$, so $\pi(r)$ is the index of the r -th highest ranked bank. For each $B_{\pi(r)}$, set $x_{\pi(r)} = 1$*

if either $\sigma_{\pi(r)} = 0$ or if

$$\sum_{r'=1}^r \sigma_{\pi(r')} (\ell_{\pi(r')} - \theta y_{\pi(r')}) \leq \sum_{r'=1}^N (1 - \sigma_{\pi(r')}) \theta y_{\pi(r')}. \quad (28)$$

In words, for any ranking of banks, the greedy algorithm goes through them sequentially, saving the highest-ranked shocked banks until liquidity runs out (and saving all not-shocked banks as well).³⁷

We now show that an exponential network can implement the greedy algorithm:

Proposition 14 (Exponential networks implement the greedy algorithm). *Let banks be ordered by the ranking π in Definition 14: $i = \pi(i)$ and suppose inequality (28) is strict for all σ and r . There exists a threshold s^* such that the exponential network $\alpha \mathbf{F}$ with base $s < s^*$ implements the outcome obtained by the greedy algorithm when α is sufficiently large.*

This result embodies the idea that the exponential network, like the greedy algorithm, prioritizes what banks should be saved. The result does not depend on the priority ranking given by π .

But the efficiency of the outcome does depend on the priority ranking. The next definition helps define a useful one:

Definition 15 (Profitability index). *The profitability index of a bank B_i is the ratio of the benefit to the cost of avoiding liquidation when it is shocked:*

$$\text{PI}_i := \frac{\Delta_i}{\ell_i - \theta y_i}. \quad (29)$$

The profitability index here is akin to the eponymous ratio in capital budgeting, namely the ratio of the payoff to costs. The greedy algorithm corresponds to the rule prescribed for

³⁷The knapsack problem and the greedy algorithm have recently appeared elsewhere in the financial networks literature: Jackson and Pernoud (2024) study a model in which, if banks are connected in a star network, the optimal ex post bailout policy solves a knapsack problem, and they discuss a greedy algorithm as an approximate way to solve it. In our model, the planner's problem is always a knapsack problem—it is independent of the network—and the exponential network turns out to be an exact implementation of the greedy algorithm in equilibrium.

investment under capital constraints and mutually exclusive projects: Undertake those with the highest profitability indices.

When banks are ranked by their profitability indices, the greedy algorithm, and hence an exponential network, is optimal in some circumstances and nearly optimal in many others:

Corollary 2 (Optimality of exponential network with common costs). *Suppose all shocked banks have the same liquidity shortfall: $\ell_i - \theta y_i = \ell_j - \theta y_j$ for all i and j and inequality 28 never binds for any σ and r . Let banks be ordered by their profitability indices, $PI_i \geq PI_j$ for $i \leq j$. An exponential network solves the planner's problem in Definition 12.*

Given Proposition 14, the result says that if the cost of saving every bank is the same, then saving those with the highest benefit delivers the optimum.

The algorithm need not be optimal, but it is nearly optimal if the liquidated banks are small:

Corollary 3 (Approximate optimality of exponential network with small banks). *As in Lemma 8, let i^* be the index of the first liquidated bank. An exponential network can deliver a deadweight loss within Δ_{i^*} of the solution to the planner's problem in Definition 12.*

This result, which follows from an application of linear programming to the knapsack problem in Dantzig (1957), implies that the relative inefficiency of the greedy algorithm is small if banks are small, in particular if the deadweight loss of liquidating a single bank, namely B_{i^*} , is small. The result also has an analogy in capital budgeting heuristics: If you use your entire budget, then doing those investments with highest profitability is optimal. In general, the risk is that, due to indivisibility, doing those investments might leave some of your budget unused. That risk is small if investments are small. Although the greedy algorithm could still be far from optimal when Δ_{i^*} is large, limiting results elsewhere in that literature suggest that the greedy algorithm is close to optimal on average (see Calvin and Leung (2003)).

Overall, the results here add support to our finding that exponential networks implement a robust policy. The (approximate) efficiency of the priority rule is not specific to the case

in which all banks have the same size of assets in place or liquidity shocks.

8 Further Extensions

Under our baseline assumptions (i) liquidation is always inefficient, (ii) default absent liquidation is costless, and (iii) banks' pledgeability θ is given. Here we relax these assumptions (albeit only for specific networks). Relaxing the first two, we show how to choose debt levels to balance the benefit of high debt in providing liquidity to avoid inefficient liquidation per the baseline with the cost of inducing excessive continuation/default included here. Relaxing the third, we find a further benefit of high debt: It leads banks to choose their pledgeability efficiently.

8.1 Risky Assets and Too Few Liquidations

So far, we assumed that y was sufficiently large that liquidation was always inefficient. Now we assume that y can have any value (but is the same for all banks). Thus it is efficient for all shocked banks to be liquidated if y is low but not if it is high. Here we denote the threshold below which liquidation is efficient by y^* and we show how to choose debt levels in a complete network to implement the efficient liquidation policy.³⁸

We consider a complete network with debt levels F , focusing on the case in which $S\ell < N\theta y$ for all $y > y^*$ (so, in principle, no bank need be liquidated: $L^* = 0$ in Lemma 6). As the network is symmetric, each shocked/not-shocked bank makes and receives the same payments; we index all shocked banks' payments by s and not-shocked banks' by n . From equation (7), the payment to each type is a pro rata share of the repayment made by all other banks of that type and by all banks of the other type. Hence the equilibrium equations

³⁸If paying ℓ represents a physical cost, then $(1 - \theta)y^* = \ell$, i.e. the cost of liquidation equals the cost of continuation. If it includes a transfer to unmodeled creditors (like AOT's "outside obligation" v), then $y^* < \ell$. Working with a general y^* allows us to stay agnostic on the interpretation of ℓ .

for any y are:

$$\begin{cases} R_{s\Rightarrow} = \left[\min \left\{ F, \theta y - \ell + \frac{1}{N-1} \left((S-1)R_{s\Rightarrow} + (N-S)R_{n\Rightarrow} \right) \right\} \right]^+, \\ R_{n\Rightarrow} = \left[\min \left\{ F, \theta y + \frac{1}{N-1} \left(SR_{s\Rightarrow} + (N-S-1)R_{n\Rightarrow} \right) \right\} \right]^+. \end{cases} \quad (30)$$

Solving gives the equilibrium repayments:

$$\begin{cases} R_{s\Rightarrow} = \left[F - \frac{N-1}{N-S}(\ell - \theta y) \right]^+, \\ R_{n\Rightarrow} = F, \end{cases} \quad (31)$$

with shocked banks being liquidated whenever $F < \frac{N-1}{N-S}(\ell - \theta y)$. Thus the efficient outcome is implemented—banks are liquidated if $y < y^*$ but not if $y \geq y^*$ —by setting $F = \frac{N-1}{N-S}(\ell - \theta y^*)$.

Intuitively, no matter the value of y , high debts allow banks to raise liquidity. In the baseline, that is only a good thing. Here, it can be bad. But an appropriately chosen debt level implements the efficient outcome no matter the value of y .

8.2 Costly Default

So far, we assumed that, while liquidation entailed a deadweight loss, default was just a transfer. Thus we found that higher debts always (weakly) increased efficiency (in the sense of Proposition 2). An alternative notion of efficiency could minimize defaults as well as liquidations. Here we show that an exponential network can achieve this goal if the debt levels are not too high, albeit only in an example.

Here we return to the three-bank exponential network in the example in Section 5 and replace the network $\mathbf{F}$ in equation (18) with $\alpha\mathbf{F}$, so increasing α increases indebtedness. Recall that the parameters are such that with two shocked banks the constrained efficient number of liquidations is one. As shocked banks always default (see equation (5)), the

constrained efficient number of defaults is two. In the example in Section 5, we achieve the efficient number of liquidations, but not of defaults (all three banks default). Here we do both.

We achieve both goals by reducing the debt. To see that, set $\alpha = 5/8$ and observe that the clearing vector equilibrium payment is $\mathbf{R}_{\rightarrow} = (\frac{1}{40}, 0, \frac{15}{16})$: B_1 is not liquidated and B_3 does not default. (But reducing the debt too much undermines the flexibility to avoid liquidation. If debt is too low, say $\alpha = 1/2$, then the clearing vector is $\mathbf{R}_{\rightarrow} = (0, 0, \frac{3}{4})$: Both B_1 and B_2 are liquidated.)

Overall, debt should be high enough to allow banks to raise liquidity, but, as in Section 8.1, it should not be too high, this time because that would lead to unnecessary costly defaults.

8.3 Endogenous Pledgeability

So far, we assumed that pledgeability θ was exogenous. Now we let each bank choose its own θ_i at a private cost $c(\theta_i)$, after the network is in place but before shocks are realized.³⁹ The concern is that coinsurance crowds out self-insurance: A bank might choose to be illiquid and free ride on its counterparty. Here we show that it need not, albeit only in a two-bank example.

We consider two banks with reciprocal debts $F_{1 \rightarrow 2} = F_{2 \rightarrow 1} = F \geq \ell$, each shocked independently with probability $p \in (0, 1)$. We denote the cost of increasing pledgeability by c , with $c(0) = 0$, $c', c'' > 0$. Write $\theta^a := \ell/y$ for the pledgeability a bank needs to meet a shock unaided; it is what a bank would choose on its own under the assumption that $c(\theta^a) \leq p(y - \ell)$, which we maintain throughout.

The network changes what each bank needs. When only one bank is shocked, it can pledge its counterparty's assets as well as its own, so it survives whenever $\theta_1 + \theta_2 \geq \theta^a$: The pair need hold only what one bank would have held alone. When both are shocked, nothing

³⁹Per footnote 11, the choice can be read as how much cash to hold, with c capturing the forgone return on the assets it replaces.

flows between them, and each survives only on its own θ_i . The candidates are therefore $(0, 0)$, shared provision $(\theta^a/2, \theta^a/2)$, concentrated provision $(0, \theta^a)$, and full self insurance (θ^a, θ^a) ; which is optimal turns on how convex c is. But whichever it is, banks choose it:

Proposition 15 (Endogenous pledgeability). *If $c(\theta^a) \leq p(y - \ell)$, then every weakly socially optimal pledgeability profile (i.e. every maximizer of expected surplus net of pledgeability costs) is a Nash equilibrium.*

The reason is that a bank's pledgeability reaches its counterparty only in the states in which the bank is not shocked itself, so a bank that cuts back loses its own protection first. At the shared profile, cutting to zero leaves the pair short of θ^a , so the deviator is liquidated when it alone is shocked; the inequality making shared provision socially better than $(0, 0)$ is exactly the one making that deviation privately unprofitable. Because each bank both provides and receives insurance, the externality appears on both sides of the planner's comparison and cancels.

As in Section 4, this relies on debts being high: It is the claim on a counterparty that makes its liquidity available. (The one fragile profile is the concentrated one, in which a single bank bears the cost while both enjoy the protection. It survives only because our maintained condition makes that bank willing to hold θ^a in autarky anyway—concentration is privately sustainable exactly when it is not really a sacrifice.)

9 Conclusion

We revisit systemic risk in financial networks, allowing existing interbank debts to be diluted with new debt, as they often can be in practice. In the model, a network of dilutable debts provides insurance among banks, and the right one implements the constrained-efficient transfers of liquidity. Even absent state-contingent contracts, dilution makes repayments state contingent: A shocked bank borrows against its interbank claims, and its counterparties bear the cost.

Dilutability can thus reverse the usual reading of what makes a system fragile. High gross debts and close ties spread distress when debts cannot be diluted, but liquidity when they can. Dilution can then substitute for policies that impose losses on creditors administratively (e.g., Bernard, Capponi, and Stiglitz (2022)): It implements a “backdoor bail-in,” in which debtors avoid insolvency by shifting the cost of distress onto diluted creditors.

That does not mean there is no case for regulation. True, given the right exposures, the bail-in requires no intervention. But those exposures need not emerge from bilateral contracting: Banks do not internalize the insurance they provide to the rest of the system, so they have an incentive to deviate from the socially efficient exposures. Policy aimed at the exposures banks build up before a crisis may thus substitute for bail-outs during one.

Nor should regulation flatten exposures or restrict dilution: It is asymmetry and dilutability that do the insuring. Policies that compress gross positions, cap the largest ones, or curtail the superpriority of new claims may trade real insurance for the appearance of safety.

A Proofs

A.1 Proof of Lemma 1 (Isomorphism between benchmark and AOT)

In AOT, the clearing vector is completely described by (Lemma B2, equation (B3)):

$$\mathbf{x} = \left[\min \{ \mathbf{y}\mathbf{1}, \mathbf{Q}\mathbf{x} + \mathbf{e} + \zeta A\mathbf{1} \} \right]^+. \quad (32)$$

In the case in which long-term assets are fully destroyed in default, the $\zeta = 0$ case, the equation can be re-written as

$$\mathbf{x} = \left[\min \left\{ \mathbf{y}\mathbf{1}, \mathbf{Q}\mathbf{x} + \frac{a-v}{a-v+A} (a-v+A)\mathbf{1} - \epsilon\boldsymbol{\sigma} \right\} \right]^+, \quad (33)$$

having used that, by definition, $\mathbf{e} = \mathbf{a} - \mathbf{v} - \epsilon\boldsymbol{\sigma}$ (with no cash, AOT's $c_i \equiv 0$) and denoting the profile of shock indicator, a notation AOT do not use, by $\boldsymbol{\sigma}$, as in our baseline. This is equivalent to the equilibrium in our benchmark model, which per Definition 1 is given by

$$\mathbf{R}_{\Rightarrow} = \left[\min \{ \mathbf{F}_{\Rightarrow}, \hat{\mathbf{F}}^\top \mathbf{R}_{\Rightarrow} + \theta \mathbf{y}\mathbf{1} - \ell \boldsymbol{\sigma} \} \right]^+, \quad (34)$$

where the color coding represents the mapping between the notation in the two papers, as described in Table 1.

In both models, the banks default whenever they cannot repay the face value of their debts; hence the sets of defaulting banks coincide. Likewise, in both, all defaulting banks are liquidated; hence efficiency coincides too (see Definition 2). $\square$

A.2 Proof of Lemma 2 (Netting in benchmark)

This proof generalizes AOT's proof of their Proposition 3. The idea is to show that in the equilibrium of $\alpha\mathbf{F}$ for $\alpha > 1$, each bank's shortfall $\mathbf{F}_{\Rightarrow} - \mathbf{R}_{\Rightarrow}$ is greater than it is in the

Table 1: Notations in AOT and here.

	AOT	This paper
Face value of debt	y_{ji}	$F_{i \rightarrow j}$
Payment received	$[\mathbf{Q}\mathbf{x}]_i$	$[\hat{\mathbf{F}}^\top \mathbf{R}_{\Rightarrow}]_i$
Negative shock	$a - v - e_i$	$\ell \sigma_i$
Total assets	$a - v + A$	y
Pledgeable assets	$a - v$	θy
Non-pledgeable assets	A	$(1 - \theta)y$

equilibrium of $\mathbf{F}$ and therefore so is the number of defaults.

Lemma A.1. *Define the mapping*

$$\Psi^\alpha : \mathbf{D} \mapsto \left[\min\{\alpha \mathbf{F}_{\Rightarrow}, \hat{\mathbf{F}}^\top \mathbf{D} - \theta y \mathbf{1} + \ell \boldsymbol{\sigma}\} \right]^+. \quad (35)$$

If $\mathbf{R}_{\Rightarrow}^\alpha$ is a clearing vector of $\alpha \mathbf{F}$, then the “shortfall” $\mathbf{D}^\alpha := \alpha \mathbf{F}_{\Rightarrow} - \mathbf{R}_{\Rightarrow}^\alpha$ is a fixed point of Ψ^α .

Proof. We compute, using $\mathbf{Q} \equiv \hat{\mathbf{F}}^\top$:

$$\begin{aligned} \alpha \mathbf{F}_{\Rightarrow} - \mathbf{R}_{\Rightarrow}^\alpha &= \alpha \mathbf{F}_{\Rightarrow} - \max\{\mathbf{0}, \min\{\alpha \mathbf{F}_{\Rightarrow}, \mathbf{Q} \mathbf{R}_{\Rightarrow}^\alpha + \theta y \mathbf{1} - \ell \boldsymbol{\sigma}\}\} && \text{(Payment Eqm.)} \\ &= \min\{\alpha \mathbf{F}_{\Rightarrow}, \max\{\mathbf{0}, \alpha \mathbf{F}_{\Rightarrow} - \mathbf{Q} \mathbf{R}_{\Rightarrow}^\alpha - \theta y \mathbf{1} + \ell \boldsymbol{\sigma}\}\} && \text{(Combining)} \\ &= \left[\min\{\alpha \mathbf{F}_{\Rightarrow}, \alpha \mathbf{F}_{\Rightarrow} - \mathbf{Q} \mathbf{R}_{\Rightarrow}^\alpha - \theta y \mathbf{1} + \ell \boldsymbol{\sigma}\} \right]^+ && \text{(Interchange min/max)} \\ &= \left[\min\{\alpha \mathbf{F}_{\Rightarrow}, \mathbf{Q}(\alpha \mathbf{F}_{\Rightarrow} - \mathbf{R}_{\Rightarrow}^\alpha) - \theta y \mathbf{1} + \ell \boldsymbol{\sigma}\} \right]^+ && \text{(zero-net debt)} \end{aligned}$$

Substituting from the definitions of $\mathbf{D}^\alpha$ and Ψ^α gives the result. $\square$

Now we show that, for $\alpha > 1$, Ψ^α has a fixed point that is no smaller than $\mathbf{D}^1$.

Lemma A.2. *Let $\mathbf{D}^1$ be a fixed point of Ψ^1 and define*

$$\mathcal{H}^\alpha := \prod_{i=1}^N [D_i^1, \alpha F_{i \rightarrow}] \quad (36)$$

For $\alpha > 1$, Ψ^α maps $\mathcal{H}^\alpha$ into itself, i.e. $\Psi^\alpha(\mathcal{H}^\alpha) \subset \mathcal{H}^\alpha$.

Proof. The upper bound, i.e. that $\Psi^\alpha(\mathbf{D}^\alpha) \leq \alpha \mathbf{F}_\Rightarrow$, follows immediately from the definition of Ψ^α as a minimum.

So we need only to show the lower bound, i.e. that $\Psi^\alpha(\mathbf{D}^\alpha) \geq \mathbf{D}^1$ for all $\mathbf{D}^\alpha \in \mathcal{H}^\alpha$. We have that for $\mathbf{D}^\alpha \in \mathcal{H}^\alpha$, $\mathbf{D}^1 \leq \mathbf{D}^\alpha$ by definition of the domain. Thus we can compute:

$$\Psi^\alpha(\mathbf{D}^\alpha) = \left[\min \{ \alpha \mathbf{F}_\Rightarrow, \mathbf{Q} \mathbf{D}^\alpha - \theta y \mathbf{1} + \ell \boldsymbol{\sigma} \} \right]^+ \quad (37)$$

$$\geq \left[\min \{ \alpha \mathbf{F}_\Rightarrow, \mathbf{Q} \mathbf{D}^1 - \theta y \mathbf{1} + \ell \boldsymbol{\sigma} \} \right]^+ \quad (38)$$

$$\geq \left[\min \{ \mathbf{F}_\Rightarrow, \mathbf{Q} \mathbf{D}^1 - \theta y \mathbf{1} + \ell \boldsymbol{\sigma} \} \right]^+ \quad (39)$$

$$\equiv \Psi(\mathbf{D}^1) \equiv \mathbf{D}^1, \quad (40)$$

since $\mathbf{D}^1$ is a fixed point of Ψ^1 by definition. $\square$

Combining the two lemmata above and applying Brouwer's theorem yields a fixed point $\mathbf{D}^\alpha \in \mathcal{H}^\alpha$. Reversing the algebra in the proof of Lemma A.1 shows that this fixed point corresponds to a clearing vector of $\alpha \mathbf{F}$. Generically, it is therefore the equilibrium shortfall vector of $\alpha \mathbf{F}$. Hence $\mathbf{D}^\alpha \geq \mathbf{D}^1$, so every bank liquidated under $\mathbf{F}$ is also liquidated under $\alpha \mathbf{F}$. $\square$

A.3 Proof of Lemma 3 (Delta connectedness in benchmark)

This proof mirrors AOT's proof of their Proposition 6; we translate it to our notation, adapt it to our notion of efficiency, and add some details.

For statement (i), w.l.o.g., we write $F := F_{i \Rightarrow}$ for the ring.

Proof of statement (i). Take $\ell > N\theta y$, which is sufficient for ℓ to be sufficiently large. We show, by verification, that in equilibrium all banks are liquidated and, therefore, no network is less efficient than the ring. Assuming all banks are liquidated and letting the

single shocked bank be B_1 , w.l.o.g., the equilibrium is the solution to

$$\begin{cases} R_{1 \rightarrow 2} &= [\theta y - \ell + R_{N \rightarrow 1}]^+ \\ R_{i \rightarrow i+1} &= \theta y + R_{i-1 \rightarrow i} \quad i \in \{2, \dots, N\}, \end{cases} \quad (41)$$

with the convention that $N+1 := 1$ for indices. Solving, we have the equilibrium: $R_{1 \rightarrow 2} = 0$ and, for $i > 1$, $R_{i \rightarrow i+1} = (i-1)\theta y \leq (N-1)\theta y$, which is less than F by hypothesis. To see that there is no other equilibrium, observe that every equilibrium satisfies $R_{i \rightarrow i+1} \leq \theta y + R_{i-1 \rightarrow i}$ for $i \geq 2$, and hence $R_{N \rightarrow 1} \leq R_{1 \rightarrow 2} + (N-1)\theta y$. The shocked bank's repayment equation therefore gives

$$R_{1 \rightarrow 2} \leq [R_{1 \rightarrow 2} + N\theta y - \ell]^+. \quad (42)$$

Since $\ell > N\theta y$, this requires $R_{1 \rightarrow 2} = 0$. The remaining repayment equations then uniquely give $R_{i \rightarrow i+1} = (i-1)\theta y$ for $i > 1$. I.e., all banks are liquidated.

Proof of statement (ii). Here we show that if $\mathbf{F}$ is δ -connected and δ is small, then not all banks are liquidated and, therefore, the network is more efficient than the ring (in which they are). Let $\mathcal{B}$ be one side of a cut witnessing that $\mathbf{F}$ is δ -connected, chosen so that the single shocked bank is not in $\mathcal{B}$; if it lies in the original set, take the complement. To do so, we show two lemmata.

Lemma A.3. *Let $\mathcal{B}$ be so chosen. If $N\delta \max_k F_{k \rightarrow} < \theta y$, then*

$$\sum_{i \in \mathcal{B}} \sum_{j \in \mathcal{B}^c} R_{i \rightarrow j} < \theta y |\mathcal{B}|. \quad (43)$$

The result says that the total payment from banks in $\mathcal{B}$ to those in $\mathcal{B}^c$ is small when δ is small.

Proof. By the definition of delta-connectedness, $F_{i \rightarrow j} = \hat{F}_{i \rightarrow j} F_{i \rightarrow} \leq \delta F_{i \rightarrow}$, for all $(i, j) \in \mathcal{B} \times \mathcal{B}^c$. Thus $R_{i \rightarrow j} \leq F_{i \rightarrow j} \leq \delta F_{i \rightarrow}$, for all $i \in \mathcal{B}$ and $j \in \mathcal{B}^c$. Now summing i over $\mathcal{B}$,

summing j over $\mathcal{B}^c$, and using $\delta|\mathcal{B}^c|\max_k F_{k\rightarrow} < N\delta\max_k F_{k\rightarrow} < \theta y$, gives

$$\sum_{i \in \mathcal{B}} \sum_{j \in \mathcal{B}^c} R_{i \rightarrow j} \leq \delta|\mathcal{B}^c| \sum_{i \in \mathcal{B}} F_{i\rightarrow} \leq \delta \max_k F_{k\rightarrow} |\mathcal{B}| |\mathcal{B}^c| < \theta y |\mathcal{B}|. \quad (44)$$

□

Lemma A.4. *Maintain the assumptions that the single shocked bank is not in $\mathcal{B}$ and that $\mathbf{F}$ is δ -connected with $N\delta\max_k F_{k\rightarrow} < \theta y$. If inequality (43) holds, then not all banks in $\mathcal{B}$ are liquidated.*

Proof. Suppose, in anticipation of a contradiction, that all banks in $\mathcal{B}$ are liquidated. Thus for each B_j in $\mathcal{B}$, $R_{j\rightarrow} = R_{j\leftarrow} + \theta y$. Summing over banks in $\mathcal{B}$, we have that $\sum_{j \in \mathcal{B}} R_{j\rightarrow} = \sum_{j \in \mathcal{B}} R_{j\leftarrow} + \theta y |\mathcal{B}|$. Now just expand $R_{j\rightarrow}$ and $R_{j\leftarrow}$ into their component payments,

$$\sum_{j \in \mathcal{B}} \left(\sum_{i \in \mathcal{B}} R_{j \rightarrow i} + \sum_{i \in \mathcal{B}^c} R_{j \rightarrow i} \right) = \sum_{j \in \mathcal{B}} \left(\sum_{i \in \mathcal{B}} R_{i \rightarrow j} + \sum_{i \in \mathcal{B}^c} R_{i \rightarrow j} \right) + \theta y |\mathcal{B}|, \quad (45)$$

and cancel $\sum_{j \in \mathcal{B}} \sum_{i \in \mathcal{B}} R_{j \rightarrow i}$ to get that

$$\sum_{j \in \mathcal{B}} \sum_{i \in \mathcal{B}^c} R_{j \rightarrow i} = \sum_{j \in \mathcal{B}} \sum_{i \in \mathcal{B}^c} R_{i \rightarrow j} + \theta y |\mathcal{B}| \geq \theta y |\mathcal{B}| \quad (46)$$

This contradicts equation (43), which says the total payment from $\mathcal{B}$ to $\mathcal{B}^c$ is small. □

A.4 Proof of Lemma 4 (Default radius in benchmark)

The result is the same as AOT's. Hence we omit the proof. □

A.5 Proof of Lemma 5 (Bottleneck connectedness in benchmark)

Statement (i) follows by combining AOT's Proposition 8 with their bound relating the bottleneck parameter to harmonic distance. For statement (ii), that bound is not directly

applicable because its lower bound concerns the maximum harmonic distance over all ordered pairs, whereas the identity of the shocked bank is fixed. We therefore use a direct cut-payment argument, analogous to AOT's proof of their Proposition 6(ii).

We begin with a lemma that connects the bottleneck parameter β to the harmonic distance d :

Lemma A.5. *Let $\mathbf{F}$ be a symmetric regular financial network of N banks. The bottleneck parameter β satisfies*

$$\frac{1}{2N\beta} \leq \max_{i,k:i \neq k} d_{k \rightarrow i} \leq \frac{16}{N\beta^2}. \quad (47)$$

Proof. This is AOT's Lemma 1 (p. 580). Hence we omit the proof. $\square$

With this we proceed to the two statements of the lemma.

Proof of statement (i). For $\beta > \beta^{ST} \equiv 4\sqrt{\frac{\theta y}{NF}}$, we have, from Lemma A.5, that

$$d_{j \rightarrow i} \leq \frac{16}{N\beta^2} < \frac{16}{N(\beta^{ST})^2} = \frac{F}{\theta y} \equiv d^{ST} \quad \text{for all } i \neq j, \quad (48)$$

per the definition of d^{ST} in Lemma 4. That lemma implies that when B_j is shocked, then each B_i defaults.

Proof of statement (ii). Let $\mathcal{B}$ be a nonempty proper subset attaining the minimum in Definition 6. Since the network is symmetric, replacing $\mathcal{B}$ with $\mathcal{B}^c$ leaves the cut value unchanged, so choose the minimizing side such that the shocked bank B_j is not in $\mathcal{B}$.

Suppose, in anticipation of a contradiction, that all banks are liquidated. Every bank in $\mathcal{B}$ is not shocked and, since liquidation and default coincide in the benchmark, equation (6) implies

$$R_{i \rightarrow} - R_{i \leftarrow} = \theta y \quad \text{for every } i \in \mathcal{B}. \quad (49)$$

Summing over $\mathcal{B}$, cancelling payments internal to $\mathcal{B}$, using $R_{i \rightarrow k} \leq F_{i \rightarrow k}$, and then using

regularity and the definition of β gives

$$\theta y|\mathcal{B}| = \sum_{i \in \mathcal{B}} \sum_{k \in \mathcal{B}^c} (R_{i \rightarrow k} - R_{k \rightarrow i}) \quad (50)$$

$$\leq \sum_{i \in \mathcal{B}} \sum_{k \in \mathcal{B}^c} F_{i \rightarrow k} \quad (51)$$

$$= F\beta|\mathcal{B}||\mathcal{B}^c| \quad (52)$$

$$\leq FN\beta|\mathcal{B}|. \quad (53)$$

But $\beta < \beta_{ST} \leq \theta y/(2NF)$ makes the last expression strictly less than $\theta y|\mathcal{B}|/2$, a contradiction. Thus at least one bank in $\mathcal{B}$ is not liquidated. $\square$

A.6 Proof of Proposition 1 (Existence and uniqueness)

Existence. A payment equilibrium is a fixed point of the mapping

$$\Phi(\mathbf{R}) := \left[\min \left\{ \mathbf{F}_{\Rightarrow}, \theta y \mathbf{1} - \ell \boldsymbol{\sigma} + \hat{\mathbf{F}}^\top \mathbf{R} \right\} \right]^+ \quad (54)$$

(equation (7)). Φ maps the complete lattice $\prod_{i=1}^N [0, F_{i \Rightarrow}]$ into itself and is monotone, since $\hat{\mathbf{F}}^\top$ is entrywise non-negative and $x \mapsto [\min\{F, x\}]^+$ is nondecreasing. By Tarski's fixed point theorem, a payment equilibrium exists; moreover, there is a least payment equilibrium $\mathbf{R}_{\Rightarrow}$.

Uniqueness. Let $\mathbf{R}'_{\Rightarrow}$ be any payment equilibrium, so $\mathbf{R}'_{\Rightarrow} \geq \mathbf{R}_{\Rightarrow}$; write $\Delta R_i := R'_{i \Rightarrow} - R_{i \Rightarrow} \geq 0$ and let $\mathcal{C} := \{i : \Delta R_i > 0\}$. (Banks with $F_{i \Rightarrow} = 0$ repay zero in every equilibrium, so every bank in $\mathcal{C}$ has $F_{i \Rightarrow} > 0$.) Since, per equation (6), each bank's repayment increases at most one for one with the repayments it receives,

$$\Delta R_i \leq R'_{i \Leftarrow} - R_{i \Leftarrow} = \sum_{k \neq i} \hat{F}_{k \rightarrow i} \Delta R_k \quad \text{for every } i. \quad (55)$$

Suppose $\mathcal{C} \neq \emptyset$. Summing inequality (55) over $i \in \mathcal{C}$, using $\Delta R_k = 0$ for $k \notin \mathcal{C}$ and

$\sum_{i \neq k} \hat{F}_{k \rightarrow i} = 1$, gives

$$\sum_{i \in \mathcal{C}} \Delta R_i \leq \sum_{k \in \mathcal{C}} \Delta R_k \sum_{i \in \mathcal{C} \setminus \{k\}} \hat{F}_{k \rightarrow i} \leq \sum_{k \in \mathcal{C}} \Delta R_k. \quad (56)$$

The two ends coincide, so both inequalities bind. The second binds only if $\sum_{i \in \mathcal{C}} \hat{F}_{k \rightarrow i} = 1$ for every $k \in \mathcal{C}$, i.e. only if banks in $\mathcal{C}$ have no liabilities outside $\mathcal{C}$. The first binds only if inequality (55) binds for every $i \in \mathcal{C}$, which, given the piecewise form of equation (5)—slope zero below zero and above $F_{i \rightarrow}$, slope one in between—requires every $i \in \mathcal{C}$ to be in the middle region in both equilibria:

$$R_{i \rightarrow} = \theta y - \ell \sigma_i + R_{i \leftarrow} \geq 0 \quad \text{and} \quad R'_{i \rightarrow} = \theta y - \ell \sigma_i + R'_{i \leftarrow} \leq F_{i \rightarrow}. \quad (57)$$

Summing the first equation in condition (57) over $i \in \mathcal{C}$ and cancelling the payments within $\mathcal{C}$ (which appear on both sides, as banks in $\mathcal{C}$ pay only within $\mathcal{C}$) leaves

$$S_{\mathcal{C}} \ell = |\mathcal{C}| \theta y + \sum_{k \notin \mathcal{C}} \sum_{i \in \mathcal{C}} \hat{F}_{k \rightarrow i} R_{k \rightarrow}, \quad (58)$$

where $S_{\mathcal{C}}$ is the number of shocked banks in $\mathcal{C}$: The shocks in $\mathcal{C}$ exactly exhaust its pledgeable assets and inflows. For any fixed repayment regime outside $\mathcal{C}$, the last term on its right-hand side is pinned down by banks outside $\mathcal{C}$: Because banks in $\mathcal{C}$ have no liabilities outside $\mathcal{C}$, their resources and repayments do not enter the clearing problem on $\mathcal{C}^c$. Within each such regime, repayments outside $\mathcal{C}$ are piecewise affine and weakly increasing in θy . Hence the right-hand side of equation (58) strictly increases in θy , because it contains the term $|\mathcal{C}| \theta y$, whereas the left-hand side is fixed for fixed ℓ . There are only finitely many choices of $\mathcal{C}$ and repayment regimes, so it holds only on a lower-dimensional subset of parameters, and the payment equilibrium is generically unique.

Payoff equivalence at the knife edge. Banks outside $\mathcal{C}$ are unaffected by the multiplicity. Since no repayment differences leave $\mathcal{C}$, they receive identical payments from banks in $\mathcal{C}$; and

since repayments among banks outside $\mathcal{C}$ are identical by construction, their total inflows are the same in both equilibria. Hence their liquidation decisions and payoffs coincide. For every $i \in \mathcal{C}$, condition (57) and $\Delta R_i > 0$ imply $0 \leq R_{i\rightarrow} < R'_{i\rightarrow} \leq F_{i\rightarrow}$. Hence B_i is not liquidated in either equilibrium and obtains $(1 - \theta)y$ in both. Thus all payment equilibria induce the same liquidation decisions and payoffs. $\square$

A.7 Proof of Proposition 2 (Netting)

The proof is similar to that of Lemma 2, but simpler because we work with repayments $\mathbf{R}_{\rightarrow}$ directly instead of “shortfalls” $\mathbf{F}_{\rightarrow} - \mathbf{R}_{\rightarrow}$.

Define the mapping

$$\Phi^\alpha(\mathbf{R}) := \left[\min \{ \alpha \mathbf{F}_{\rightarrow}, \mathbf{Q}\mathbf{R} + \theta y \mathbf{1} - \ell \boldsymbol{\sigma} \} \right]^+. \quad (59)$$

Keeping in mind that $\mathbf{Q} \equiv \hat{\mathbf{F}}^\top$, the fixed point of Φ^1 is a clearing vector of $\mathbf{F}$. $\square$

Lemma A.6. *Let $\mathbf{R}_{\rightarrow}^1$ be a fixed point of Φ^1 and define*

$$\mathcal{J}^\alpha := \prod_{i=1}^N [R_{i\rightarrow}^1, \alpha F_{i\rightarrow}]. \quad (60)$$

For $\alpha > 1$, Φ^α maps $\mathcal{J}^\alpha$ into itself, i.e. that $\Phi^\alpha(\mathcal{J}^\alpha) \subset \mathcal{J}^\alpha$.

Proof. The upper bound, i.e. that $\Phi^\alpha(\mathbf{R}_{\rightarrow}^\alpha) \leq \alpha \mathbf{F}_{\rightarrow}$, follows immediately from the definition of Φ^α as a minimum.

So we need only to show the lower bound, i.e. that $\Phi^\alpha(\mathbf{R}_{\rightarrow}^\alpha) \geq \mathbf{R}_{\rightarrow}^1$. For $\mathbf{R}_{\rightarrow}^\alpha \in \mathcal{J}^\alpha$, we

can compute:

$$\Phi^\alpha(\mathbf{R}_{\Rightarrow}^\alpha) = \left[\min \{ \alpha \mathbf{F}_{\Rightarrow}, \mathbf{Q} \mathbf{R}_{\Rightarrow}^\alpha + \theta y \mathbf{1} - \ell \boldsymbol{\sigma} \} \right]^+ \quad (61)$$

$$\geq \left[\min \{ \alpha \mathbf{F}_{\Rightarrow}, \mathbf{Q} \mathbf{R}_{\Rightarrow}^1 + \theta y \mathbf{1} - \ell \boldsymbol{\sigma} \} \right]^+ \quad (62)$$

$$\geq \left[\min \{ \mathbf{F}_{\Rightarrow}, \mathbf{Q} \mathbf{R}_{\Rightarrow}^1 + \theta y \mathbf{1} - \ell \boldsymbol{\sigma} \} \right]^+ \quad (63)$$

$$\equiv \Phi^1(\mathbf{R}_{\Rightarrow}^1) \equiv \mathbf{R}_{\Rightarrow}^1, \quad (64)$$

since $\mathbf{R}_{\Rightarrow}^1$ is a fixed point of Φ by definition. □

Given the lemma, we can apply Brouwer's theorem, to conclude that for $\alpha > 1$, an equilibrium of $\alpha \mathbf{F}$ is a fixed point of a mapping on $\mathcal{J}^\alpha$. Therefore the generically unique⁴⁰ clearing vector, being in $\mathcal{J}^\alpha$, exceeds $\mathbf{R}_{\Rightarrow}^1$: All repayments are higher when α is higher, so there are fewer liquidations. □

A.8 Proof of Proposition 3 (Delta connectedness)

Note that this proof makes use of the minimum number of liquidated banks, L^* , derived in Lemma 6, even though that result comes later in the text.

Throughout we assume, w.l.o.g., that the B_1 is the not-shocked bank.

Proof of statement (i). Equation (5) and the fact that banks have zero net positions (so no shocked bank can repay in full) imply that any shocked bank ($i \geq 2$) repays

$$R_{i\Rightarrow} \equiv R_{i \rightarrow i+1} = [\theta y - \ell + R_{i\Leftarrow}]^+ \quad (65)$$

$$= [\theta y - \ell + R_{i-1 \rightarrow i}]^+, \quad (66)$$

⁴⁰Generic global uniqueness follows from Proposition 1. When the equilibrium is not unique, let $\mathbf{R}_{\Rightarrow}^1$ be the least fixed-point as per Proposition 1. The outcome-equivalence part of that proposition implies that all equilibria of each LT network induce the same liquidation set. Therefore comparison of these two fixed points suffices for the efficiency ranking.

having used the definition of the ring network (Definition 3).

The expression for $R_{i\rightarrow}$ implies that if B_{i-1} is liquidated, then B_i is too. Thus the number of banks that are not liquidated is the maximum index i for which $R_{i-1\rightarrow i} \geq \ell - \theta y$.

We can now expand the condition recursively for any B_i that is not liquidated:

$$\ell - \theta y \leq R_{i-1\rightarrow i} = \theta y - \ell + R_{i-2\rightarrow i-1} \quad (67)$$

$$= k(\theta y - \ell) + R_{i-(k+1)\rightarrow i-k} \quad (\text{for } k \in \{1, \dots, i-2\}) \quad (68)$$

$$= (i-2)(\theta y - \ell) + R_{1\rightarrow 2}. \quad (69)$$

So the number of banks that are not liquidated is

$$\max \left\{ i \leq N : i-1 \leq \frac{R_{1\rightarrow 2}}{\ell - \theta y} \right\} = \min \left\{ N, \left\lfloor 1 + \frac{R_{1\rightarrow 2}}{\ell - \theta y} \right\rfloor \right\}. \quad (70)$$

The number of liquidated banks is N minus the above:

$$L = \max \left\{ 0, \left\lfloor N - 1 - \frac{R_{1\rightarrow 2}}{\ell - \theta y} \right\rfloor \right\}. \quad (71)$$

Suppose first that $(N-1)(\ell - \theta y) > \theta y$. At least one shocked bank must be liquidated: Otherwise, iterating their repayment equations around the ring and using $R_{1\rightarrow 2} \leq \theta y + R_{N\rightarrow 1}$ would imply $(N-1)(\ell - \theta y) \leq \theta y$, a contradiction. Once one shocked bank is liquidated, every shocked bank after it is too, so B_N is liquidated and $R_{N\rightarrow 1} = 0$. Hence $R_{1\rightarrow 2} = \theta y$, since $F > \theta y$. Substituting into the preceding expression for L and comparing with equation (10) at $S = N-1$ gives

$$L = \left\lfloor N - 1 - \frac{\theta y}{\ell - \theta y} \right\rfloor = L^*. \quad (72)$$

Suppose instead that $(N-1)(\ell - \theta y) \leq \theta y$. Then the repayment vector

$$R_{1\rightarrow 2} = F \quad \text{and} \quad R_{i\rightarrow i+1} = F - (i-1)(\ell - \theta y) \quad \text{for } i \in \{2, \dots, N\} \quad (73)$$

is an equilibrium in which no bank is liquidated: The repayment equations hold for every shocked bank and $\theta y + R_{N \rightarrow 1} \geq F$ makes B_1 repay in full. The outcome-equivalence part of Proposition 1 implies the same liquidation outcome in every payment equilibrium, so $L = 0 = L^*$. In either case, the ring attains the minimum number of liquidations and is therefore most efficient among all networks.

Statement (ii). A trivial example suffices: The “linkless” network— $F_{i \rightarrow j} = 0$ for all i and j —is δ -connected for any δ ; in it, all $N - 1$ shocked banks are liquidated. Since $\ell \leq 2\theta y$ implies $L^* \leq N - 2$, that is less efficient than the ring network with $F > \theta y$, per statement (i). $\square$

A.9 Proof of Proposition 4 (Salvation radius)

We prove the two statements in turn. The arguments build on AOT’s proof of their Proposition 8 (pp. 602–603). Ours are a bit more complicated because we cannot work with the clearing vector $\mathbf{R}_{\Rightarrow}$, but have to work with the shortfall $\mathbf{D} \equiv \mathbf{F}_{\Rightarrow} - \mathbf{R}_{\Rightarrow}$ instead.

Statement (i). The proof comprises (somewhat involved) calculations using the shortfall $\mathbf{D}$, which ultimately allow us to bound the harmonic distance of the not-shocked banks to any liquidated bank.

Throughout we denote the set of defaulting banks by $\mathcal{D}$, that of liquidated banks by $\mathcal{L}$, and, hence, that of those that default but are not liquidated by $\mathcal{D} \setminus \mathcal{L}$. $\mathbf{1}$ denotes the vector of all ones and $\mathbf{I}$ the identity matrix, each of appropriate dimension determined by the context. Given the network is regular, we can write $F_{i \Rightarrow} \equiv F$.

We start with two lemmata. Each takes as its starting point the equilibrium equation for the shortfall

$$\mathbf{D} = \left[\min\{\mathbf{F}_{\Rightarrow}, \mathbf{Q}\mathbf{D} - \theta y \mathbf{1} + \ell \boldsymbol{\sigma}\} \right]^+, \quad (74)$$

which follows from setting $\alpha = 1$ in Lemma A.1. The first lemma develops the equation for banks that default but are not liquidated; the second for banks that are liquidated.

Lemma A.7. $\mathbf{D}_{\mathcal{D} \setminus \mathcal{L}} = (\mathbf{I} - \mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{D} \setminus \mathcal{L}})^{-1} (\mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{L}} \mathbf{1} F - (\theta y - \ell) \mathbf{1})$.

Proof. For banks that default, the shortfall is not zero, and for those that are not liquidated, it is less than F . Hence for B_i in $\mathcal{D} \setminus \mathcal{L}$, the second term under the “min” in equation (74) is the relevant one; writing that elementwise gives the following:

$$D_i = \sum_{k=1}^N Q_{ik} D_k - \theta y + \ell \quad (75)$$

$$= \sum_{k \in \mathcal{L}} Q_{ik} D_k + \sum_{k \in \mathcal{D} \setminus \mathcal{L}} Q_{ik} D_k + \sum_{k \notin \mathcal{D}} Q_{ik} D_k - \theta y + \ell \quad (76)$$

$$= \sum_{k \in \mathcal{L}} Q_{ik} D_k + \sum_{k \in \mathcal{D} \setminus \mathcal{L}} Q_{ik} D_k - \theta y + \ell, \quad (77)$$

having used that shortfall is zero for banks that do not default ($D_k = 0$ for $k \notin \mathcal{D}$). Rewriting the above in block matrix form, using that fact that liquidated banks repay nothing ($D_k = F$ for $k \in \mathcal{L}$), and rearranging gives:

$$\mathbf{D}_{\mathcal{D} \setminus \mathcal{L}} = \mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{D} \setminus \mathcal{L}} \mathbf{D}_{\mathcal{D} \setminus \mathcal{L}} + \mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{L}} \mathbf{D}_{\mathcal{L}} - (\theta y - \ell) \mathbf{1} \quad (78)$$

$$= \mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{D} \setminus \mathcal{L}} \mathbf{D}_{\mathcal{D} \setminus \mathcal{L}} + \mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{L}} F \mathbf{1} - (\theta y - \ell) \mathbf{1} \quad (79)$$

$$= (\mathbf{I} - \mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{D} \setminus \mathcal{L}})^{-1} (\mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{L}} F \mathbf{1} - (\theta y - \ell) \mathbf{1}), \quad (80)$$

where the last expression comes from solving for $\mathbf{D}_{\mathcal{D} \setminus \mathcal{L}}$ and rearranging. $\square$

Lemma A.8. $(\mathbf{I} + \mathbf{Q}_{\mathcal{L}, \mathcal{D} \setminus \mathcal{L}} (\mathbf{I} - \mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{D} \setminus \mathcal{L}})^{-1}) (\ell - \theta y) \mathbf{1} > \tilde{\mathbf{Q}} F \mathbf{1}$, where

$$\tilde{\mathbf{Q}} := (\mathbf{I} - \mathbf{Q}_{\mathcal{L}, \mathcal{D} \setminus \mathcal{L}} (\mathbf{I} - \mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{D} \setminus \mathcal{L}})^{-1} \mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{L}} - \mathbf{Q}_{\mathcal{L}, \mathcal{L}}). \quad (81)$$

Proof. Banks that are liquidated repay zero (equation (5)), so

$$F < \sum_{k=1}^N Q_{ik} D_k - \theta y + \ell \quad (82)$$

$$= \sum_{k \in \mathcal{L}} Q_{ik} D_k + \sum_{k \in \mathcal{D} \setminus \mathcal{L}} Q_{ik} D_k + \sum_{k \notin \mathcal{D}} Q_{ik} D_k - \theta y + \ell \quad (83)$$

$$= \sum_{k \in \mathcal{L}} Q_{ik} D_k + \sum_{k \in \mathcal{D} \setminus \mathcal{L}} Q_{ik} D_k - \theta y + \ell \quad (84)$$

$$= \sum_{k \in \mathcal{L}} Q_{ik} F + \sum_{k \in \mathcal{D} \setminus \mathcal{L}} Q_{ik} D_k - \theta y + \ell, \quad (85)$$

having used that shortfall is zero for banks that do not default ($D_k = 0$ for $k \notin \mathcal{D}$) and F for those that are liquidated ($D_k = F$ for $k \in \mathcal{L}$). The above can be re-written in block-matrix notation, $F\mathbf{1} < \mathbf{Q}_{\mathcal{L}, \mathcal{D} \setminus \mathcal{L}} \mathbf{D}_{\mathcal{D} \setminus \mathcal{L}} + \mathbf{Q}_{\mathcal{L}, \mathcal{L}} F\mathbf{1} - (\theta y - \ell)\mathbf{1}$, so the expression for $\mathbf{D}_{\mathcal{D} \setminus \mathcal{L}}$ from Lemma A.7 can be substituted in to get:

$$\mathbf{Q}_{\mathcal{L}, \mathcal{D} \setminus \mathcal{L}} (\mathbf{I} - \mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{D} \setminus \mathcal{L}})^{-1} (\mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{L}} F\mathbf{1} - (\theta y - \ell)\mathbf{1}) + \mathbf{Q}_{\mathcal{L}, \mathcal{L}} F\mathbf{1} - (\theta y - \ell)\mathbf{1} > F\mathbf{1}. \quad (86)$$

Rearranging the above gives the expression in the lemma. $\square$

Now we compute a bound on the harmonic distance d . First we use the definition of d (Definition 5) to write in block matrix form:

$$\begin{cases} \mathbf{d}_{j \rightarrow \mathcal{L}} = \mathbf{1} + \mathbf{Q}_{\mathcal{L}, \mathcal{L}} \mathbf{d}_{j \rightarrow \mathcal{L}} + \mathbf{Q}_{\mathcal{L}, \mathcal{D} \setminus \mathcal{L}} \mathbf{d}_{j \rightarrow \mathcal{D} \setminus \mathcal{L}}, \\ \mathbf{d}_{j \rightarrow \mathcal{D} \setminus \mathcal{L}} = \mathbf{1} + \mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{L}} \mathbf{d}_{j \rightarrow \mathcal{L}} + \mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{D} \setminus \mathcal{L}} \mathbf{d}_{j \rightarrow \mathcal{D} \setminus \mathcal{L}}, \end{cases} \quad (87)$$

where $\mathbf{d}_{j \rightarrow \mathcal{L}}$ and $\mathbf{d}_{j \rightarrow \mathcal{D} \setminus \mathcal{L}}$ are vectors that capture the harmonic distances from B_j to each of (i) the liquidated and (ii) the defaulting but not liquidated banks, respectively (cf. equations (B19) and (B20) in AOT). (NB: As, by hypothesis, B_j is the only not-shocked bank, there are no additional terms to not defaulting banks.)

Solving for the system in equation (87)—solving for $\mathbf{d}_{j \rightarrow \mathcal{D} \setminus \mathcal{L}}$ in the second equation and substituting it into the first—gives

$$\mathbf{d}_{j \rightarrow \mathcal{L}} = \mathbf{1} + \mathbf{Q}_{\mathcal{L}, \mathcal{L}} \mathbf{d}_{j \rightarrow \mathcal{L}} + \mathbf{Q}_{\mathcal{L}, \mathcal{D} \setminus \mathcal{L}} (\mathbf{I} - \mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{D} \setminus \mathcal{L}})^{-1} (\mathbf{1} + \mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{L}} \mathbf{d}_{j \rightarrow \mathcal{L}}) \quad (88)$$

or, given the definition of $\tilde{\mathbf{Q}}$ in equation (81), $\tilde{\mathbf{Q}} \mathbf{d}_{j \rightarrow \mathcal{L}} = (\mathbf{I} + \mathbf{Q}_{\mathcal{L}, \mathcal{D} \setminus \mathcal{L}} (\mathbf{I} - \mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{D} \setminus \mathcal{L}})^{-1}) \mathbf{1}$.

From here, we can use Lemma A.8 to write

$$\tilde{\mathbf{Q}} \mathbf{d}_{j \rightarrow \mathcal{L}} > \tilde{\mathbf{Q}} \frac{F}{\ell - \theta y} \mathbf{1}. \quad (89)$$

As $\tilde{\mathbf{Q}}$ is non-singular M-matrix, its inverse exists and is elementwise non-negative,⁴¹ this says that if B_i is liquidated, then

$$d_{j \rightarrow i} \geq \frac{F}{\ell - \theta y} \equiv d^{LT}, \quad (90)$$

per the definition of d^{LT} in the proposition. That is the desired result.

Statement (ii). As, by hypothesis, no bank is liquidated, we have that for any B_i $D_i < F_{i \rightleftharpoons}$, from the definition of the shortfall $\mathbf{D}$. Thus, from the equilibrium equation for the shortfall (Lemma A.1 with $\alpha = 1$) and the observation that no shocked bank repays in full, $D_i > 0$ (equation (5) given the assumption that banks have zero net positions), we have

$$D_i = \sum_{k \neq i} Q_{ik} D_k + \ell - \theta y, \quad (91)$$

for any B_i for $i \neq j$, where, remember, B_j is the not-shocked bank. Dividing both sides of equation (91) by $\ell - \theta y$ says that $D_i/(\ell - \theta y)$ solves $x_i = 1 + \sum_{k \neq i} Q_{ik} x_k$ for all i . By the

⁴¹The result follows Theorem 2 of Plemmons (1977) and exercise 5.8 of Berman and Plemmons (1979, p. 159) given that $\tilde{\mathbf{Q}}$ is the Schur complement of the non-singular M -matrix

$$\begin{bmatrix} \mathbf{I} - \mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{D} \setminus \mathcal{L}} & -\mathbf{Q}_{\mathcal{D} \setminus \mathcal{L}, \mathcal{L}} \\ -\mathbf{Q}_{\mathcal{L}, \mathcal{D} \setminus \mathcal{L}} & \mathbf{I} - \mathbf{Q}_{\mathcal{L}, \mathcal{L}} \end{bmatrix}.$$

definition (and uniqueness) of the harmonic distance, that implies that

$$d_{j \rightarrow i} = \frac{D_i}{\ell - \theta y}. \quad (92)$$

As $D_i < F$ by hypothesis, $d_{j \rightarrow i} < F/(\ell - \theta y) \equiv d^{LT}$, as desired. $\square$

A.10 Proof of Proposition 5 (Bottleneck connectedness)

We prove the two statements of the proposition sequentially. Statement (i) relies on Lemma A.5 above (which, recall, depends only on the network structure, not the maturity of debt despite being stated within the benchmark of Section 3). Statement (ii) instead uses a direct cut-payment argument.

Proof of statement (i). For $\beta > \beta^{LT} \equiv 4\sqrt{\frac{\ell - \theta y}{NF}}$, we have, from Lemma A.5, that

$$d_{j \rightarrow i} \leq \frac{16}{N\beta^2} < \frac{16}{N(\beta^{LT})^2} = \frac{F}{\ell - \theta y} \equiv d^{LT}, \quad (93)$$

per the definition of d^{LT} in Proposition 4. That result implies that when B_j is not shocked, then B_i is not liquidated.

Proof of statement (ii). Let $\mathcal{B}$ be a nonempty proper subset attaining the minimum in Definition 6. Since the network is symmetric, replacing $\mathcal{B}$ with $\mathcal{B}^c$ leaves the cut value unchanged, so choose the minimizing side such that the unique not-shocked bank B_j is not in $\mathcal{B}$. Thus every bank in $\mathcal{B}$ is shocked.

Suppose, in anticipation of a contradiction, that no bank is liquidated. A shocked bank cannot repay in full: By symmetry and regularity, it receives at most its total interbank claims F , while $\ell > \theta y$ means that repaying its total liabilities F would require an incoming payment strictly greater than F . Hence every bank in $\mathcal{B}$ defaults without being liquidated, and equation (6) implies

$$R_{i \Leftarrow} - R_{i \rightarrow} = \ell - \theta y \quad \text{for every } i \in \mathcal{B}. \quad (94)$$

Summing over $\mathcal{B}$, cancelling payments internal to $\mathcal{B}$, using $R_{k \rightarrow i} \leq F_{k \rightarrow i}$, and then using symmetry, regularity, and the definition of β gives

$$(\ell - \theta y)|\mathcal{B}| = \sum_{i \in \mathcal{B}} \sum_{k \in \mathcal{B}^c} (R_{k \rightarrow i} - R_{i \rightarrow k}) \quad (95)$$

$$\leq \sum_{i \in \mathcal{B}} \sum_{k \in \mathcal{B}^c} F_{k \rightarrow i} \quad (96)$$

$$= F\beta|\mathcal{B}||\mathcal{B}^c| \quad (97)$$

$$\leq FN\beta|\mathcal{B}|. \quad (98)$$

But $\beta < \beta_{LT} \leq (\ell - \theta y)/(2NF)$ makes the last expression strictly less than $(\ell - \theta y)|\mathcal{B}|/2$, a contradiction. Thus at least one bank in $\mathcal{B}$ is liquidated. $\square$

A.11 Proof of Lemma 6 (Constrained efficiency)

The argument is in the text. $\square$

A.12 Proof of Lemma 7 (High debt mutualizes assets)

We allow bank heterogeneity by indexing ℓ and y with subscripts since the lemma is invoked in the proof of Proposition 14. We show that if a not-shocked bank, say B_i , makes net payment less than θy_i no matter how high α is, it is impossible that another bank, say B_j , is liquidated. From equation (5) with proper indexing, the not-shocked bank's net payment is

$$R_{i \rightarrow} - R_{i \leftarrow} = \begin{cases} \theta y_i & \text{if defaults,} \\ \alpha F_{i \rightarrow} - R_{i \leftarrow} & \text{otherwise.} \end{cases} \quad (99)$$

Since the network is connected and has zero net positions, it is strongly connected,⁴²

⁴²To see this, contract each strongly connected component into one node. If there were more than one component, the resulting finite directed acyclic graph would have a source component $\mathcal{C}$. No liabilities enter $\mathcal{C}$, while zero net positions, summed over its banks, imply that total liabilities leaving $\mathcal{C}$ equal total liabilities entering it and hence are also zero. This contradicts (weak) connectivity unless $\mathcal{C}$ is the whole network.

meaning there is a directed path $i = i_0 \rightarrow i_1 \rightarrow \dots \rightarrow i_m = j$ with $F_{i_t \rightarrow i_{t+1}} > 0$ on every edge.

Next we show along this path, if $R_{i_{t-1} \rightarrow i_t} \geq \alpha a_t - b_t$ for some $a_t > 0$ and finite $b_t \geq 0$ independent of α , then $R_{i_t \rightarrow i_{t+1}} \geq \alpha a_{t+1} - b_{t+1}$, for some $a_{t+1} > 0$ and finite $b_{t+1} \geq 0$ independent of α . To see this, set $a_1 := F_{i_0 \rightarrow i_1}$ and $b_1 := 0$, and recursively

$$\begin{aligned} a_{t+1} &:= \hat{F}_{i_t \rightarrow i_{t+1}} \min \{F_{i_t \rightarrow i_{t+1}}, a_t\} > 0, \\ b_{t+1} &:= \hat{F}_{i_t \rightarrow i_{t+1}} [b_t + \ell_{i_t} \sigma_{i_t} - \theta_{i_t} y_{i_t}]^+ < \infty. \end{aligned}$$

Since B_{i_0} repays in full, $R_{i_0 \rightarrow i_1} = \alpha F_{i_0 \rightarrow i_1}$. We now apply the repayment equation successively along the path. If $R_{i_{t-1} \rightarrow i_t} \geq \alpha a_t - b_t$ for some $a_t > 0$ and finite $b_t \geq 0$, then

$$\begin{aligned} R_{i_t \rightarrow i_{t+1}} &= \hat{F}_{i_t \rightarrow i_{t+1}} R_{i_t \rightarrow i_{t+1}} \\ &\geq \hat{F}_{i_t \rightarrow i_{t+1}} [\min \{\alpha F_{i_t \rightarrow i_{t+1}}, \alpha a_t - b_t + \theta_{i_t} y_{i_t} - \ell_{i_t} \sigma_{i_t}\}]^+ \quad \text{since } R_{i_{t-1} \rightarrow i_t} \geq \alpha a_t - b_t \\ &\geq \alpha a_{t+1} - b_{t+1} \quad \text{since } [\min \{\alpha u, \alpha v - c\}]^+ \geq \alpha \min \{u, v\} - [c]^+ \text{ for } u, v \geq 0. \end{aligned}$$

Induction therefore implies $R_{i_{m-1} \rightarrow j} \geq \alpha a_m - b_m$. For sufficiently large α , this exceeds B_j 's finite liquidity deficit, contradicting its liquidation. Taking the largest of these finitely many thresholds over ordered pairs (i, j) and shock profiles σ proves that, if any bank is liquidated, every not-shocked bank makes net payment θy_i .

A.13 Proof of Corollary 1 (First best with high debt for small shocks)

Suppose (in anticipation of a contradiction) that at least one bank is liquidated. From Lemma 7 and equation (5), we know that for each B_i the net payment is

$$R_{i\rightarrow} - R_{i\leftarrow} \begin{cases} = \theta y - \ell \sigma_i & \text{if } B_i \text{ is not liquidated,} \\ > \theta y - \ell \sigma_i & \text{if } B_i \text{ is liquidated.} \end{cases} \quad (100)$$

Combining market clearing (equation (4)) with the expression above, we have that

$$0 = \sum (R_{i\rightarrow} - R_{i\leftarrow}) > \sum (\theta y - \ell \sigma_i) = N\theta y - S\ell, \quad (101)$$

contradicting the hypothesis that $N\theta y > S\ell$. Therefore no bank can be liquidated, as desired. $\square$

A.14 Proof of Lemma 8 (Controlling relative payments to liquidated banks)

Use the pro rata condition that $R_{i\rightarrow j} = \hat{F}_{i\rightarrow j} R_{i\rightarrow}$ for any j (including $j = i^*$), by Definitions 8 and 9, after relabeling banks according to their common hierarchy,

$$R_{i\rightarrow j} = \hat{F}_{i\rightarrow j} R_{i\rightarrow} \quad (102)$$

$$\leq s^{j-i^*} \hat{F}_{i\rightarrow i^*} R_{i\rightarrow} \quad (103)$$

$$= s^{j-i^*} R_{i\rightarrow i^*}. \quad (104)$$

Since B_{i^*} is liquidated, $R_{i^* \rightarrow j} = 0$. Therefore,

$$R_{j \Leftarrow} = \sum_{i \neq j, i^*} R_{i \rightarrow j} \quad (105)$$

$$\leq s^{j-i^*} \sum_{i \neq j, i^*} R_{i \rightarrow i^*} \quad (106)$$

$$\leq s^{j-i^*} R_{i^* \Leftarrow}. \quad (107)$$

□

A.15 Proof of Lemma 9 (Controlling total payments to liquidated banks)

Since liquidated banks make zero repayments, $R_{i \rightarrow} = 0$ for $i \in \mathcal{L}$, equation (5) implies that each receives payment $R_{i \Leftarrow} < \ell - \theta y$. Applying this to B_{i^*} , the highest-ranked liquidated bank, and using Lemma 8, we have the following:

$$\sum_{i \in \mathcal{L}} R_{i \Leftarrow} \leq \sum_{i \in \mathcal{L}} s^{i-i^*} R_{i^* \Leftarrow} \quad (108)$$

$$\leq R_{i^* \Leftarrow} \sum_{i=0}^{\infty} s^i \quad (109)$$

$$= R_{i^* \Leftarrow} \frac{1}{1-s} \quad (110)$$

$$< \frac{\ell - \theta y}{1-s}. \quad (111)$$

□

A.16 Proof of Proposition 6 (Efficiency of exponential networks)

As the exponential network is connected, we know from Lemma 6 and Corollary 1 that if $L^* = 0$ then no bank is liquidated as long as α is high. Hence we focus on the case in which at least one bank is liquidated in the constrained-efficient outcome, $L^* \geq 1$.

Recall that in this case it suffices to show the following:

- (i) Each bank that is not liquidated makes the maximum net payment it can without being liquidated, $R_{i\Rightarrow} - R_{i\Leftarrow} = \theta y - \ell \sigma_i$, for B_i not liquidated.
- (ii) The banks that are liquidated receive a total net payment that would be insufficient to save any one of them, $-\sum_{i \in \mathcal{L}} (R_{i\Rightarrow} - R_{i\Leftarrow}) < \ell - \theta y$.

The first property follows from Lemma 7 and equation (5).

The second property follows from two steps. The first is to use Lemma 9 and the definition of s^* to bound the liquidated banks' net payment in terms of L^* :

$$-\sum_{i \in \mathcal{L}} (R_{i\Rightarrow} - R_{i\Leftarrow}) < \frac{\ell - \theta y}{1 - s} \quad (112)$$

$$\leq \frac{\ell - \theta y}{1 - s^*} \quad (113)$$

$$= N\theta y - S\ell + (1 + L^*)(\ell - \theta y). \quad (114)$$

The second step is to use market clearing, $\sum_{i=1}^N (R_{i\Rightarrow} - R_{i\Leftarrow}) = 0$ by equation (4), to write the LHS above in terms of the number of liquidated banks L , using that (i) not-shocked banks make net payment θy (by Lemma 7) and (ii) shocked, not-liquidated banks make net payment $\theta y - \ell$ (by equation (5)):

$$-\sum_{i \in \mathcal{L}} (R_{i\Rightarrow} - R_{i\Leftarrow}) = \sum_{i \in \mathcal{L}^c} (R_{i\Rightarrow} - R_{i\Leftarrow}) \quad (115)$$

$$= \sum_{i \in \mathcal{L}^c : \sigma_i=0} (R_{i\Rightarrow} - R_{i\Leftarrow}) + \sum_{i \in \mathcal{L}^c : \sigma_i=1} (R_{i\Rightarrow} - R_{i\Leftarrow}) \quad (116)$$

$$= (N - S)\theta y + (S - L)(\theta y - \ell). \quad (117)$$

Combining this with the bound in equation (114) and canceling terms says $L < 1 + L^*$. As L and L^* are integers, and $L^* \leq L$ by Lemma 6, it must be that $L = L^*$.

(The assumption that $\frac{S\ell - N\theta y}{\ell - \theta y}$ not be an integer was required for $s^* > 0$ and thus for the exponential network with base $s < s^*$ to be well defined.) $\square$

A.17 Proof of Proposition 7 (Approximate efficiency of exponential networks)

With the weaker notion of efficiency, we need to show only that the banks that are liquidated receive a total net payment insufficient to save any *two* of them, $-\sum_{i \in \mathcal{L}} (R_{i\Rightarrow} - R_{i\Leftarrow}) < 2(\ell - \theta y)$. Given the proof of Proposition 6, that is all we need to show. As in that proof, it follows Lemma 9 along with the definitions of $s^*(= 1/2)$ and L^* :

$$-\sum_{i \in \mathcal{L}} (R_{i\Rightarrow} - R_{i\Leftarrow}) < \frac{\ell - \theta y}{1 - s^*} \quad (118)$$

$$= 2(\ell - \theta y) \quad (119)$$

as desired. $\square$

A.18 Proof of Lemma 10 (Inefficiency of other networks)

Suppose (in anticipation of a contradiction) that a fully connected network $\mathbf{F}$ achieves constrained efficiency, i.e. that the number of liquidated banks is $L^* = \frac{S\ell - N\theta y}{\ell - \theta y}$, having used the assumptions that $\frac{S\ell - N\theta y}{\ell - \theta y}$ is an integer and that $S\ell > N\theta y$ in conjunction with the definition of L^* (equation (10)).

As each bank that is not shocked pays at most θy and each that is shocked but not liquidated pays exactly $\theta y - \ell\sigma_i$ (equation (5)), we can use market clearing to bound the

total payment to the liquidated banks as follows:

$$-\sum_{i \in \mathcal{L}} (R_{i \Rightarrow} - R_{i \Leftarrow}) = \sum_{i \notin \mathcal{L}} (R_{i \Rightarrow} - R_{i \Leftarrow}) \quad (120)$$

$$\leq \sum_{i \notin \mathcal{L}} (\theta y - \sigma_i \ell) \quad (121)$$

$$= (N - S)\theta y + (S - L^*)(\theta y - \ell) \quad (122)$$

$$= 0. \quad (123)$$

I.e. liquidated banks receive no (positive) payment.

But shocked, not-liquidated banks must receive positive payment (otherwise they would be liquidated by equation (5)). Given the hypothesis that the network is fully connected, that contradicts the assumption that payments are pro rata (equation (3)): Any bank that has debt to a not-liquidated bank must have debt to a liquidated bank too and it cannot make a positive payment to one but not the other. $\square$

A.19 Proof of Proposition 8 (Exponential not pairwise stable)

Since α is sufficiently large, all not-shocked banks default and obtain a payoff of $(1 - \theta)y$, and if exactly one of B_{N-1} and B_N is shocked, it is liquidated. We show they can deviate to large mutual debts which help them avoid liquidation when one of them is shocked and do not affect their payoff when both or neither of them is shocked.

Let $O_i = \alpha \sum_{k \notin \{N-1, N\}} F_{i \rightarrow k}$ for $i \in \{N-1, N\}$ denote the aggregate liabilities of B_i to banks outside the pair. Consider a deviation $F'_{N-1 \rightarrow N} = F'_{N \rightarrow N-1} = F$ such that

$$F > \max_{i \in \{N-1, N\}} \left\{ \alpha F_{N-1 \rightarrow N}, \alpha F_{N \rightarrow N-1}, \theta y, \frac{(\ell - \theta y)O_i}{2\theta y - \ell} \right\}.$$

Consider first a state in which only one of the two banks, B_i , is shocked and the other bank B_j is not shocked. Since $L^* \geq 1$, B_i is liquidated and obtains a payoff of zero.

After the deviation, B_j 's total liabilities are $F + O_j > \theta y$. And it pays at least $\frac{F}{F+O_j}\theta y > \ell - \theta y$ to B_i where the strict inequality follows from $F(2\theta y - \ell) > (\ell - \theta y)O_j$. Thus, B_i avoids liquidation. The not-shocked bank B_j is not worse off in this state since it obtains at least $(1 - \theta)y$. The same argument applies symmetrically when B_j is shocked and B_i is not.

If neither bank in the pair is shocked, both obtain at least the original payoff $(1 - \theta)y$.

When both are shocked, if $L^* \geq 2$, both are liquidated, so neither can be made worse off. Suppose instead $L^* = 1$. Then B_{N-1} is saved and B_N is liquidated in the original equilibrium. We claim that B_{N-1} still survives after the deviation. If B_{N-1} were liquidated, then B_N 's only incoming payments would come from banks outside the pair. Given the exponential structure, $R'_{N\Leftarrow} \leq sR'_{N-1\Leftarrow} < \ell - \theta y$, so B_N would be liquidated too. But once B_{N-1} and B_N are both liquidated, neither of them makes any payment, and the debts between every other pair remain unchanged, so the equilibrium payment vector is also the solution to the payment equation before deviation. But this contradicts that all payment equilibria induce the same set of liquidated banks (see the proof of Proposition 1): The constructed payment vector is an equilibrium of the pre-deviation network in which B_{N-1} is liquidated, whereas in the original equilibrium it is not.

Since the shock distribution assigns positive probability to both states in which only one of B_{N-1} and B_N is shocked, their expected payoffs are strictly improved.

A.20 Proof of Proposition 9 (Star pairwise stable)

We first present the equilibrium of the star network for different values of F .

Lemma A.9. *Suppose $1 < S < N$ and $\ell - \theta y \leq (N - S)\theta y$ so that the system has enough liquidity to save at least one bank.⁴³ Let $\bar{F} := \frac{N}{S}\theta y$, $\bar{F}^c := \frac{N\theta y - \ell}{S-1}$, $\underline{F} := \frac{\ell - \theta y}{N-S}$ and $\underline{F} := \frac{\theta y}{S}$.*

- *When only peripheral banks are shocked, all of them are liquidated. The core bank defaults if $F > \underline{F}$ and the not-shocked peripheral banks default if $F > \bar{F}$.*

⁴³We only need this for the case when the core bank is shocked.

- When the core bank is also shocked, all the shocked peripheral banks are liquidated. The core bank is liquidated if $F < \underline{F}^c$. The not-shocked banks default when $F > \bar{F}^c$.

Proof. The liquidation of the peripheral banks comes from the symmetry argument given insufficient liquidity in the network.

The not-shocked peripheral banks default only if the core defaults on them and the default is large enough. When the core is not shocked, it defaults iff its payment at the verge of default is weakly less than its liabilities

$$R_{1\Rightarrow} = (N - S - 1)F + \theta y \leq (N - 1)F \iff F \geq \underline{F}. \quad (124)$$

The peripheral banks default iff

$$\frac{R_{1\Rightarrow}}{N - 1} + \theta y < F \iff F > \bar{F}. \quad (125)$$

When the core is shocked, it is liquidated if it doesn't receive enough liquidity at the verge of liquidation

$$(N - S) \min\{F, \theta y\} < \ell - \theta y \iff F < \underline{F}^c \quad \text{given} \quad \ell - \theta y \leq (N - S)\theta y. \quad (126)$$

As long as the not-shocked peripheral banks repay in full, and hence at the verge of their default, the core pays $R_{1\Rightarrow} = (N - S)F + \theta y - \ell$ once it survives. The not-shocked peripheral banks default if

$$\frac{R_{1\Rightarrow}}{N - 1} + \theta y < F \iff F > \bar{F}^c. \quad (127)$$

□

Without deviation, given that $F \geq \underline{F}^c > \underline{F}$, the core bank is never liquidated, but always defaults and obtains $(1 - \theta)y$ in every shock state. We show that the core bank can never obtain a higher payoff in every state, no matter the deviation, by showing that the maximum

residual pledgeable assets (i.e., pledgeable assets minus liabilities assuming the core does not default) are negative. Let's consider a deviation between the core bank B_1 and a peripheral bank B_2 to $F_{1 \rightarrow 2}$ and $F_{2 \rightarrow 1}$. Denote the liquidity deficit $D := \ell - \theta y$. The assumptions $S\ell > (2N - S)\theta y$ and $S > 1$ imply

$$\frac{D}{N - S} > \frac{2\theta y}{S} \geq \frac{\theta y}{S - 1}. \quad (128)$$

- Suppose first that the core is not shocked and B_2 is shocked. If the core were solvent, it would pay $F_{1 \rightarrow 2}$ to B_2 and F to each of the other $N - 2$ peripheral banks. Hence the core's residual pledgeable assets are at most

$$\begin{aligned} & \theta y + (F_{1 \rightarrow 2} - D)^+ + (S - 1)(F - D)^+ + (N - S - 1)F - F_{1 \rightarrow 2} - (N - 2)F. \\ & = \theta y + (F_{1 \rightarrow 2} - D)^+ - F_{1 \rightarrow 2} + (S - 1)[(F - D)^+ - F] \end{aligned}$$

When $F \leq D$, the expression is smaller than $\theta y - (S - 1)F$ ⁴⁴ which is negative since $F > \frac{\theta y}{S - 1}$; when $F > D$, the expression is smaller than $\theta y - (S - 1)D$ which is smaller than 0 since $D > \frac{(N - S)\theta y}{S - 1} \geq \frac{\theta y}{S - 1}$.

- Next suppose both the core and B_2 are not shocked (only possible when $N - S \geq 2$). Since B_2 can repay at most $\theta y + F_{1 \rightarrow 2}$, the core's residual pledgeable assets are at most

$$\begin{aligned} & \theta y + (\theta y + F_{1 \rightarrow 2}) + S(F - D)^+ + (N - S - 2)F - F_{1 \rightarrow 2} - (N - 2)F. \\ & = 2\theta y + S[(F - D)^+ - F] \end{aligned}$$

When $F \geq D$, the expression equals $2\theta y - SD < 0$ since $D > \frac{2\theta y}{S}(N - S) \geq \frac{2\theta y}{S}$; when $F < D$, the expression equals $2\theta y - SF < 0$ since $F \geq \frac{D}{N - S} > \frac{2\theta y}{S}$.

- Now suppose both the core and B_2 are shocked. The core's residual pledgeable assets

⁴⁴We used the (self-evident) auxiliary inequality $(x - y)^+ - x < 0 \quad \forall x, y > 0$ here and below.

are at most

$$\begin{aligned} & \theta y - \ell + (F_{1 \rightarrow 2} - D)^+ + (S - 2)(F - D)^+ + (N - S)F - F_{1 \rightarrow 2} - (N - 2)F = \\ & = -D + [(F_{1 \rightarrow 2} - D)^+ - F_{1 \rightarrow 2}] + (S - 2)[(F - D)^+ - F] < -D < 0. \end{aligned}$$

- Finally, suppose the core is shocked and B_2 is not shocked. The core's residual pledgeable assets are at most

$$\begin{aligned} & \theta y - \ell + (\theta y + F_{1 \rightarrow 2}) + (S - 1)(F - D)^+ + (N - S - 1)F - F_{1 \rightarrow 2} - (N - 2)F. \\ & = 2\theta y - \ell + (S - 1)[(F - D)^+ - F] \\ & = \{2\theta y + S[(F - D)^+ - F]\} - \{\ell + [(F - D)^+ - F]\} \end{aligned}$$

We have proven that the term in the first curly bracket is negative and that the term in the second, $\ell + [(F - D)^+ - F] \geq \ell - D = \theta y > 0$, is positive, so the entire expression is negative. Intuitively, being directly hit is worse than being defaulted on by a shocked bank.

Thus, after any possible core-periphery deviation, the core bank's payoff is at most its original payoff $(1 - \theta)y$ in every shock state, and hence in expectation.

A.21 Proof of Proposition 9' (Star still pairwise stable)

The large-shock condition, $N\theta y < (N - 1)\ell$, is equivalent to $\ell - \theta y > \theta y / (N - 1)$. Since $F = \ell - \theta y < \theta y / (N - 2) \leq \theta y$, a not-shocked peripheral can repay $\ell - \theta y$ in full, which saves the shocked core. Since $\ell - \theta y > \theta y / (N - 1)$, the core cannot save all $N - 1$ shocked peripheral banks in the original star but it defaults. The core obtains $(1 - \theta)y$ in every state without deviation. A peripheral bank, say, B_2 , obtains a positive payoff only when it is the unique unshocked peripheral; in that state it pays $\ell - \theta y$ to the core, receives no repayment from the core, and obtains $y - (\ell - \theta y)$.

By symmetry, consider a deviation $F_{1 \rightarrow 2}$ and $F_{2 \rightarrow 1}$ between B_1 and B_2 . We will show that i) the core bank B_1 can only strictly benefit in the state when it is not shocked, and it must be that $F_{1 \rightarrow 2} < \ell - \theta y$, and thus, B_2 remains liquidated whenever shocked; ii) for B_1 to strictly benefit in expectation, it must not be liquidated in the state when B_2 is not shocked, and it requires $F_{2 \rightarrow 1} \geq \ell - \theta y$, but then B_2 cannot strictly gain in that state.

First, suppose B_1 is the unique not-shocked bank. It is the only state in which B_1 can strictly gain.⁴⁵ A non-deviating shocked peripheral can receive at most $\ell - \theta y$ from the core; even if it receives exactly $\ell - \theta y$, it only covers its liquidity shortfall and has no resources left to repay the core. Hence, if the core is solvent, its residual pledgeable assets are at most

$$\theta y + (F_{1 \rightarrow 2} - (\ell - \theta y))^+ - F_{1 \rightarrow 2} - (N - 2)(\ell - \theta y).$$

If $F_{1 \rightarrow 2} \geq \ell - \theta y$, this upper bound is $\theta y - (N - 1)(\ell - \theta y) < 0$. If $F_{1 \rightarrow 2} < \ell - \theta y$, it is $\theta y - F_{1 \rightarrow 2} - (N - 2)(\ell - \theta y)$. Thus the core can be strictly better than $(1 - \theta)y$ in this state only if

$$F_{1 \rightarrow 2} < \theta y - (N - 2)(\ell - \theta y).$$

This inequality implies $F_{1 \rightarrow 2} < \ell - \theta y$ because $\ell - \theta y > \theta y / (N - 1)$.

Second, suppose B_2 is the unique not-shocked peripheral. If $F_{2 \rightarrow 1} < \ell - \theta y$, then the core is liquidated and loses its baseline payoff $(1 - \theta)y$. And it is no less than its possible gain $\theta y - (N - 2)(\ell - \theta y)$. Hence, a deviation with $F_{2 \rightarrow 1} < \ell - \theta y$ cannot strictly raise the core's expected payoff given uniform distribution. Suppose instead $F_{2 \rightarrow 1} \geq \ell - \theta y$. If the core is liquidated in this state, then B_2 receives no payment from the core; B_2 either repays at least $\ell - \theta y$ and obtains a payoff at most $y - (\ell - \theta y)$, or defaults and obtains $(1 - \theta)y < y - (\ell - \theta y)$.

If the core is not liquidated, let $R_{1 \rightarrow 2}$ and $R_{2 \rightarrow 1}$ be the realized payments between B_1 and B_2 .

⁴⁵After deviation, B_1 need not have zero net position when shocked and may not default even when shocked. Nevertheless, only the unique not-shocked peripheral supplies external pledgeable liquidity, bounded by θy ; payments recycled through shocked peripherals cannot create positive net resources. For B_1 to become solvent, this external liquidity must cover its liquidity shortfall and all $N - 2$ unchanged liabilities, requiring at least $(N - 1)(\ell - \theta y) > \theta y$, which is impossible. Hence B_1 cannot strictly gain whenever it is shocked.

Since B_1 is shocked, any payment it makes to interbank creditors must come from resources left after covering the liquidity shortfall $\ell - \theta y$. The other shocked peripheral banks have no resources to repay the core. Therefore, $R_{1 \rightarrow 2} \leq R_{2 \rightarrow 1} - (\ell - \theta y)$. If B_2 is solvent, its payoff is

$$y + R_{1 \rightarrow 2} - R_{2 \rightarrow 1} \leq y - (\ell - \theta y).$$

If B_2 defaults, it obtains $(1 - \theta)y < y - (\ell - \theta y)$. Thus, B_2 cannot strictly gain in the state in which it is the unique not-shocked peripheral when $F_{2 \rightarrow 1} \geq \ell - \theta y$.

Combining the two points above, any deviation that strictly benefits B_1 leaves B_2 weakly worse off in every shock state. Hence no bilateral deviation can strictly benefit both banks in expectation.

Finally, we show B_1 can avoid default. Take the deviation $F_{1 \rightarrow 2} = 0$ and $F_{2 \rightarrow 1} = \ell - \theta y$. In the state in which all peripheral banks are shocked, the core owes only $\ell - \theta y$ to each of $B_3, \dots, B_N$, repays all these liabilities in full, has residual pledgeable assets $\theta y - (N - 2)(\ell - \theta y) > 0$, and obtains more than $(1 - \theta)y$. B_2 is weakly indifferent across states. Thus pairwise stability does not come from the core always defaulting after deviations.

A.22 Proof of Proposition 10 (Uniform complete pairwise stable)

Let

$$\bar{F} = \frac{N}{N - 2}\theta y,$$

and fix $F \geq \bar{F}$, a shock distribution G , and a bilateral deviation between B_1 and B_2 . In any shock state with $S = |\sigma| \in \{1, \dots, N - 1\}$, the condition $\ell - \theta y > (N - 1)\theta y$ implies $L^* = S$: even all $N - S$ not-shocked banks together have at most $(N - S)\theta y \leq (N - 1)\theta y < \ell - \theta y$, which is not enough to save one shocked bank. Hence every shocked bank is liquidated before and after the deviation.

Relabeling the two deviating banks if necessary, suppose $F_{1 \rightarrow 2} \geq F_{2 \rightarrow 1}$ after the deviation.

In the original uniform complete network, every not-shocked bank defaults in such a state.

Indeed, its incoming payment is at most $(N - S - 1)F$, while its liabilities are $(N - 1)F$, and $(N - 1)F > \theta y + (N - S - 1)F$ because $SF \geq F \geq \bar{F} > \theta y$. Thus every not-shocked bank's net payment is θy , and every shocked bank's payoff is zero.

Now suppose exactly one deviating bank, say B_1 , is shocked. Since B_1 is liquidated, B_1 cannot strictly gain. We show that B_2 cannot strictly gain either. The deviation most favorable to B_2 in this state sets B_2 's liability to B_1 equal to zero; B_1 's liability to B_2 is irrelevant because B_1 is liquidated. Let U be the set of not-shocked banks outside the deviating pair, and let $m = |U| = N - S - 1$. If $m > 0$, let $\bar{R}$ be the largest total repayment among banks in U . A bank in U receives at most F from B_2 and at most $(m - 1)\bar{R}/(N - 1)$ from the other banks in U . Hence

$$\bar{R} \leq \theta y + F + \frac{m - 1}{N - 1}\bar{R}, \quad \text{so} \quad \frac{\bar{R}}{N - 1} \leq \frac{\theta y + F}{S + 1}.$$

Thus B_2 's incoming payment from U is at most $m(\theta y + F)/(S + 1)$. If B_2 defaults, its net payment is θy , as in the original network. If B_2 is solvent, its net payment is at least

$$(N - 2)F - \frac{N - S - 1}{S + 1}(\theta y + F).$$

This lower bound is at least θy because

$$(N - 2)F - \frac{N - S - 1}{S + 1}(\theta y + F) - \theta y = \frac{[S(N - 1) - 1]F - N\theta y}{S + 1} \geq 0,$$

where the last inequality follows from $F \geq \bar{F} = N\theta y/(N - 2)$, since $S(N - 1) - 1 \geq N - 2$. Hence B_2 cannot strictly gain. The same argument applies when B_2 is shocked and B_1 is not.

It remains to consider states in which both deviating banks have the same shock status. If both are shocked and at least one bank is not shocked, the argument above shows that both are liquidated. If all banks are shocked, aggregate payment clearing likewise rules out

any non-liquidated bank, since such a bank would require a positive net inflow $\ell - \theta y$ but there is no source of positive net outflow. Thus B_1 cannot gain when both deviating banks are shocked. Suppose instead that both are not shocked and that S banks outside the pair are shocked. If B_1 defaults after the deviation, its payoff is $(1 - \theta)y$, equal to its original payoff when $S \geq 1$ and below its original payoff y when $S = 0$. If B_1 is solvent, it receives at most $F_{2 \rightarrow 1}$ from B_2 and at most F from each of the $N - S - 2$ not-shocked banks outside the pair. Since its total liabilities are $(N - 2)F + F_{1 \rightarrow 2}$, its net payment is at least

$$(N - 2)F + F_{1 \rightarrow 2} - [(N - S - 2)F + F_{2 \rightarrow 1}] = SF + F_{1 \rightarrow 2} - F_{2 \rightarrow 1}.$$

For $S \geq 1$, this is at least $F > \theta y$, so B_1 's payoff is no greater than its original payoff $(1 - \theta)y$. For $S = 0$, it is at least $F_{1 \rightarrow 2} - F_{2 \rightarrow 1} \geq 0$, so B_1 's payoff is no greater than its original payoff y . Thus B_1 is weakly worse off in every shock state. Therefore the deviation cannot strictly increase both banks' expected payoffs under any G .

A.23 Proof of Proposition 11 (Star inefficient)

Since the system doesn't have enough liquidity to save all shocked banks, when the core bank is not shocked, all shocked banks are peripheral, and they are all liquidated by symmetry. When the core bank is shocked, it is not possible for both the core and all peripheral banks to avoid liquidation. The core can avoid liquidation if it receives enough liquidity from the $(N - S)$ not-shocked peripheral banks, but given the low interbank debt $F < (\ell - \theta y)/(N - S)$, the maximum liquidity the core bank can receive is $(N - S)F < \ell - \theta y$. So it cannot avoid liquidation. Both scenarios are not constrained efficient when $L^* < S$ by definition.

A.24 Proof of Proposition 12 (Uniform complete inefficient)

Since there is not enough liquidity to save all shocked banks given $S\ell > N\theta y$, all shocked banks are liquidated by symmetry and equilibrium uniqueness (Proposition 1).

A.25 Proof of Proposition 13 (The planner's problem is the knapsack problem)

We prove each implication in turn.

(i) $\hat{\mathbf{t}}$ solves PP $\implies \hat{\mathbf{x}}$ solves KP. We show that $\hat{\mathbf{x}}$ is feasible and optimal in turn.

- $\hat{\mathbf{x}}$ is feasible. Observe that, since, by equation (24), $\hat{t}_i \geq -\theta y_i$, $\hat{x}_i \sigma_i (\ell_i - \theta y_i) \leq (1 - \sigma_i) \theta y_i + \hat{t}_i$. Thus, using the liquidity conservation constraint (equation (25)), we have that

$$\sum_{i=1}^N \hat{x}_i \sigma_i (\ell_i - \theta y_i) \leq \sum_{i=1}^N \left((1 - \sigma_i) \theta y_i + \hat{t}_i \right) \leq \sum_{i=1}^N (1 - \sigma_i) \theta y_i, \quad (129)$$

confirming that $\hat{\mathbf{x}}$ is feasible.

- $\hat{\mathbf{x}}$ is optimal. Suppose, in anticipation of a contradiction, that $\hat{\mathbf{x}}$ is not optimal, i.e. that there is a feasible $\hat{\mathbf{x}}'$ that yields lower deadweight loss (equation (26)).

We now show that such an $\hat{\mathbf{x}}'$ cannot exist, because, if it does, $\hat{\mathbf{t}}$ cannot be a solution to PP. Specifically, $\hat{\mathbf{t}}'$ with $\hat{t}'_i := \sigma_i \hat{x}'_i (\ell_i - \theta y_i) - (1 - \sigma_i) \theta y_i$ is feasible and yields a lower objective:

– Feasibility:

- * We show that $\hat{\mathbf{t}}'$ satisfies the liquidity constraint (24) for $\sigma_i = 0$ and $\sigma_i = 1$ in turn: If $\sigma_i = 0$, then $\hat{t}'_i = -\theta y_i = \min\{\sigma_i \ell_i - \theta y_i, 0\}$ and if $\sigma_i = 1$, then $\hat{t}'_i = \hat{x}'_i (\ell_i - \theta y_i) \geq \min\{\sigma_i \ell_i - \theta y_i, 0\}$.
- * We show that $\hat{\mathbf{t}}'$ satisfies the liquidity conservation constraint (25) as

$$\sum_{i=1}^N \hat{t}'_i = \sum_{i=1}^N \left(\sigma_i \hat{x}'_i (\ell_i - \theta y_i) - (1 - \sigma_i) \theta y_i \right) \leq 0, \quad (130)$$

since $\hat{\mathbf{x}}'$ must satisfy the constraint (27) by its definition as a solution to KP.

– Optimality: Observe, from the definitions of $\hat{x}'_i$ and $\hat{t}_i$ that

$$\sum_{i=1}^N \mathbb{1}_{\{\theta y_i + \hat{t}'_i < \ell_i \sigma_i\}} \Delta_i \leq \sum_{i=1}^N (1 - \hat{x}'_i) \Delta_i < \sum_{i=1}^N (1 - \hat{x}_i) \Delta_i = \sum_{i=1}^N \mathbb{1}_{\{\theta y_i + \hat{t}_i < \ell_i \sigma_i\}} \Delta_i, \quad (131)$$

contradicting the optimality of $\hat{\mathbf{t}}$.

Therefore $\hat{\mathbf{x}}$ solves KP.

(ii) $\check{\mathbf{x}}$ solves KP $\implies \check{\mathbf{t}}$ solves PP. We show that $\check{\mathbf{t}}$ is feasible and optimal in turn.

– $\check{\mathbf{t}}$ is feasible. Observe that $\check{t}_i$ satisfies B_i 's liquidity constraint (24) by construction:

$$\check{t}_i = \sigma_i \check{x}_i (\ell_i - \theta y_i) - (1 - \sigma_i) \theta y_i \geq \min\{\sigma_i \ell_i - \theta y_i, 0\}.$$

– $\check{\mathbf{t}}$ is optimal. Suppose, in anticipation of a contradiction, that $\check{\mathbf{t}}$ is not optimal, i.e. that there is a feasible $\check{\mathbf{t}}'$ that yields lower deadweight loss (equation (23)).

We now show that such an $\check{\mathbf{t}}'$ cannot exist because, if it does, $\check{\mathbf{x}}$ cannot be a solution to KP. Specifically, $\check{\mathbf{x}}'$ with $\check{x}'_i := \mathbb{1}_{\{\theta y_i + \check{t}'_i - \ell_i \sigma_i \geq 0\}}$ is feasible and yields a lower objective:

* Feasibility: We show that $\check{x}'_i$ satisfies the liquidity conservation constraint (27)

as

$$\sum_{i=1}^N \check{x}'_i \sigma_i (\ell_i - \theta y_i) \leq \sum_{i=1}^N \left((1 - \sigma_i) \theta y_i + \check{t}'_i \right) \leq \sum_{i=1}^N (1 - \sigma_i) \theta y_i, \quad (132)$$

since $\check{\mathbf{t}}'$ must satisfy the constraint (25) by its definition as a solution to PP.

* Optimality: Observe, from the definitions of $\check{x}'_i$ and $\check{t}_i$ that

$$\sum_{i=1}^N (1 - \check{x}'_i) \Delta_i \leq \sum_{i=1}^N \mathbb{1}_{\{\theta y_i + \check{t}'_i < \ell_i \sigma_i\}} \Delta_i < \sum_{i=1}^N \mathbb{1}_{\{\theta y_i + \check{t}_i < \ell_i \sigma_i\}} \Delta_i \leq \sum_{i=1}^N (1 - \check{x}_i) \Delta_i, \quad (133)$$

contradicting the optimality of $\check{\mathbf{x}}$.

Therefore $\check{\mathbf{t}}$ solves PP. □

A.26 Proof of Proposition 14 (Exponential networks implement the greedy algorithm)

First note that Lemma 7, Lemma 8, and Lemma 9 hold with heterogeneous banks. Their proofs are essentially unchanged, with y and ℓ replaced with y_i and ℓ_i everywhere, except in the last line of the proof of Lemma 9, when they are replaced by y_{i^*} and ℓ_{i^*} . We apply these results freely throughout the proof.

Now we prove the result in three steps.

Step 1: **For all i such that $\sigma_i = 0$, $x_i = 1$.** This is immediate from equation (5), which implies that the bank is not liquidated if it is not shocked.

Step 2: **Existence of critical index.** Now show that there exists a critical index i^* such that a shocked bank B_i is liquidated if and only if $i \geq i^*$. Suppose, in anticipation of a contradiction, that, to the contrary, there are shocked B_i and B_j with $i < j$ such that $x_i = 0$ and $x_j = 1$. Then, by equation (5), it must be that $R_{i\Leftarrow} < \ell_i - \theta y_i$ and $R_{j\Leftarrow} \geq \ell_j - \theta y_j$ and, therefore, by Lemma 8, that

$$\ell_j - \theta y_j \leq R_{j\Leftarrow} \leq s^{j-i} R_{i\Leftarrow} < s^{j-i} (\ell_i - \theta y_i). \quad (134)$$

The inequality is violated if $s < \sqrt[j-i]{(\ell_j - \theta y_j)/(\ell_i - \theta y_i)}$, a contradiction.

Step 3: **Same critical index.** We now show that the critical index delivered by the exponential network coincides with that delivered by the greedy algorithm (for every state σ) or, equivalently, that i^* satisfies the following two inequalities:

$$\sum_{i=1}^{i^*} \sigma_i (\ell_i - \theta y_i) > \sum_{i=1}^N (1 - \sigma_i) \theta y_i \quad (135)$$

and

$$\sum_{i=1}^{i^*-1} \sigma_i (\ell_i - \theta y_i) \leq \sum_{i=1}^N (1 - \sigma_i) \theta y_i. \quad (136)$$

The second is immediate, as it is implied by the aggregate liquidity constraint, which holds strictly by hypothesis. To prove the first, we invoke Lemma 7, which implies that, as long as debts are sufficiently high, each not-shocked bank pays θy_i in net payment, so conservation of liquidity requires:

$$\sum_{i=1}^N (1 - \sigma_i) \theta y_i = - \sum_{i: \sigma_i=0} (R_{i\Leftarrow} - R_{i\Rightarrow}) \quad (137)$$

$$= \sum_{i: \sigma_i=1} (R_{i\Leftarrow} - R_{i\Rightarrow}) \quad (138)$$

$$= \sum_{i \in \mathcal{L}^c: \sigma_i=1} (R_{i\Leftarrow} - R_{i\Rightarrow}) + \sum_{i \in \mathcal{L}} (R_{i\Leftarrow} - R_{i\Rightarrow}) \quad (139)$$

$$\leq \sum_{i \in \mathcal{L}^c: \sigma_i=1} (R_{i\Leftarrow} - R_{i\Rightarrow}) + \sum_{i \in \mathcal{L}} R_{i\Leftarrow} \quad (140)$$

$$< \sum_{i \in \mathcal{L}^c: \sigma_i=1} (R_{i\Leftarrow} - R_{i\Rightarrow}) + \frac{R_{i^* \Leftarrow}}{1-s} \quad (141)$$

$$\leq \sum_{i \in \mathcal{L}^c: \sigma_i=1} (\ell_i - \theta y_i) + \frac{\ell_{i^*} - \theta y_{i^*}}{1-s} \quad (142)$$

$$= \sum_{i=1}^{i^*} \sigma_i (\ell_i - \theta y_i) + \frac{s}{1-s} (\ell_{i^*} - \theta y_{i^*}) \quad (143)$$

having used Lemma 9 to bound the sum over liquidated banks. Letting s be sufficiently small and recalling that the inequality must be strict by assumption gives the result. $\square$

A.27 Proof of Corollary 2 (Optimality of exponential network with common costs)

The result is immediate from the definitions of equivalence of the planner's problem to the knapsack problem and of the greedy algorithm to the exponential network; see Proposition 13, Definition 13, and Proposition 14. (See also the discussion following the statement of the corollary.) $\square$

A.28 Proof of Corollary 3 (Approximate optimality of exponential network with small banks)

The result follows immediately from the so-called Dantzig bound (Dantzig (1957)), which we state as a lemma:

Lemma A.10. *Suppose, w.l.o.g., that banks are ordered by their profitability indices ($PI_i \geq PI_j$ for $i \leq j$) and let $\tilde{\mathbf{x}}$ be a solution of KP (Definition 13). We have that*

$$\sum_{i=1}^N \sigma_i \tilde{x}_i \Delta_i \leq \sum_{i=1}^{i^*-1} \sigma_i \Delta_i + \Delta_{i^*}. \quad (144)$$

Proof. See Martello and Toth (1990), Theorem 2.1. □

The result implies that the difference between the objective at the optimum (represented by the LHS of equation (144)) and at the greedy algorithm's approximation of it (represented by the sum on the RHS) is at most Δ_{i^*} , as desired. □

A.29 Proof of Proposition 15 (Endogenous pledgeability)

We prove the result in three steps: We first characterize which banks are liquidated as a function of the profile (θ_1, θ_2) , which reduces the planner's problem to a comparison of four candidate profiles; we then compare those profiles; and we finally check that no bank wants to deviate from any of them. Throughout, we label banks w.l.o.g. so that the concentrated profile is $(0, \theta^a)$ rather than its permutation.

Liquidation and candidate profiles. Suppose exactly one bank, say B_1 , is shocked. Since $F \geq \ell$, it can raise $\theta_1 y + \min\{F, \theta_2 y\}$, so it survives if and only if $\theta_1 + \theta_2 \geq \theta^a$. Suppose instead both banks are shocked. A bank with $\theta_i \geq \theta^a$ meets its shock unaided and repays $\theta_i y - \ell$; a bank with $\theta_i < \theta^a$ receives nothing from a liquidated counterparty and so, by equation (5), is itself liquidated unless its counterparty survives and repays enough. Hence both banks survive if and only if $\theta_1 + \theta_2 \geq 2\theta^a$, and otherwise only a bank with $\theta_i \geq \theta^a$ does.

Up to terms independent of pledgeability, the planner maximizes

$$-(y - \ell) \mathbb{E}[\#\text{liquidated banks}] - c(\theta_1) - c(\theta_2). \quad (145)$$

By the previous step, profiles fall into four classes by liquidation outcome, and within each class the planner takes the cheapest. If $\theta_1 + \theta_2 < \theta^a$, no bank is ever saved, and $(0, 0)$ is cheapest. If $\theta_1 + \theta_2 \geq \theta^a$ with $\theta_i < \theta^a$ for both banks, shocked banks are saved only in single-shock states, and strict convexity makes the equal split $(\theta^a/2, \theta^a/2)$ uniquely cheapest. If $\theta_i \geq \theta^a$ for one bank and $\theta_1 + \theta_2 < 2\theta^a$, one bank is also saved in the double-shock state, and $(0, \theta^a)$ is uniquely cheapest. If $\theta_1 + \theta_2 \geq 2\theta^a$, both banks are always saved, and strict convexity again makes (θ^a, θ^a) uniquely cheapest, ruling out $(0, 2\theta^a)$ in particular. It is therefore enough to compare these four profiles.

Social optimum. Taking $(0, 0)$ as the baseline, we have the following expression for the incremental surpluses:

$$\text{incremental surplus} = \begin{cases} 2p(1-p)(y - \ell) - 2c(\theta^a/2) & \text{for } (\theta^a/2, \theta^a/2), \\ [2p(1-p) + p^2](y - \ell) - c(\theta^a) & \text{for } (0, \theta^a) \\ 2p(y - \ell) - 2c(\theta^a) & \text{for } (\theta^a, \theta^a) \end{cases} \quad (146)$$

The maintained condition $c(\theta^a) \leq p(y - \ell)$ makes the second at least $p(1-p)(y - \ell) > 0$, so $(0, 0)$ is not socially optimal. Pairwise comparison of the other three gives the following classification:

- (i) (θ^a, θ^a) is weakly socially optimal if and only if $c(\theta^a) \leq p^2(y - \ell)$.
- (ii) $(0, \theta^a)$ is weakly socially optimal if and only if $c(\theta^a) \geq p^2(y - \ell)$ and $c(\theta^a) - 2c(\theta^a/2) \leq p^2(y - \ell)$.
- (iii) $(\theta^a/2, \theta^a/2)$ is weakly socially optimal if and only if $c(\theta^a) - 2c(\theta^a/2) \geq p^2(y - \ell)$.

Each displayed inequality is the comparison with $(0, \theta^a)$; the remaining comparison, between (θ^a, θ^a) and $(\theta^a/2, \theta^a/2)$, follows from it in each case, since $c(\theta^a/2) \geq 0$ implies $c(\theta^a) - c(\theta^a/2) \leq c(\theta^a)$ in (i) and $c(\theta^a) - c(\theta^a/2) \geq c(\theta^a) - 2c(\theta^a/2)$ in (iii). These cases exhaust the parameters satisfying the maintained condition.

Best response/equilibrium. We check each social optimum in turn.

- (i) At (θ^a, θ^a) , suppose B_i cuts to $\theta < \theta^a$. The pair still holds at least θ^a , so B_i keeps its single-shock protection, but the pair holds less than $2\theta^a$ and $\theta < \theta^a$, so B_i is liquidated when both banks are shocked, losing $p^2(y - \ell)$. The saving $c(\theta^a) - c(\theta)$ is largest at $\theta = 0$, and condition (i) rules that out. Raising pledgeability gains B_i nothing, as it is already saved in every state.
- (ii) At $(0, \theta^a)$, suppose B_1 raises to θ . If $\theta < \theta^a$ it remains liquidated when both banks are shocked, so it gains nothing; if $\theta \geq \theta^a$ it gains $p^2(y - \ell)$ at a cost of at least $c(\theta^a)$, which condition (ii) rules out. Suppose instead B_2 cuts to $\theta < \theta^a$. The pair then holds less than θ^a , so B_2 is liquidated whenever it is shocked, losing $p(y - \ell)$ and saving at most $c(\theta^a)$; the maintained condition rules that out. Raising its pledgeability further gains B_2 nothing, as it is already saved whenever it is shocked.
- (iii) At $(\theta^a/2, \theta^a/2)$, suppose B_i cuts to $\theta < \theta^a/2$. The pair then holds less than θ^a , so B_i is liquidated in the state in which it alone is shocked, losing $p(1 - p)(y - \ell)$ and saving at most $c(\theta^a/2)$. Condition (iii) and the maintained condition give

$$c(\theta^a/2) \leq \frac{c(\theta^a) - p^2(y - \ell)}{2} \leq \frac{p(1 - p)(y - \ell)}{2}, \quad (147)$$

so the deviation is unprofitable. Suppose instead B_i raises its pledgeability to θ . It gains only if it survives when both banks are shocked, which requires $\theta \geq \theta^a$ and hence costs

$$c(\theta) - c(\theta^a/2) \geq c(\theta^a) - c(\theta^a/2) \geq c(\theta^a) - 2c(\theta^a/2) \geq p^2(y - \ell), \quad (148)$$

which is the gain.

Thus every weakly socially optimal profile is a Nash equilibrium. $\square$

B Ring Example

Here we show that the ring network is not constrained efficient with four banks. We keep $y = 2$ and $\theta = 1/2$ but increase the shock to $\ell = 12/5$, so that no single bank's liquidity covers a shortfall, $\theta y = 1 < \ell - \theta y = 7/5$, although pooling two banks' liquidity does, $\ell - \theta y \leq 2\theta y$. With $S = 2$ banks shocked, exactly one bank is liquidated in the constrained-efficient outcome, $L^* = 1$ (Lemma 6).

In the ring, each B_i owes F to B_{i+1} (indices modulo four) and nothing to the others:

$$\mathbf{F} = F \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}. \quad (149)$$

Whether the ring is efficient depends on where the shocks fall.

Suppose first that the shocked banks are adjacent, say B_1 and B_2 . The clearing vector solves

$$\begin{cases} R_{1\rightarrow} = \max \left\{ 0, \min \left\{ -\frac{7}{5} + R_{4\rightarrow}, F \right\} \right\}, \\ R_{2\rightarrow} = \max \left\{ 0, \min \left\{ -\frac{7}{5} + R_{1\rightarrow}, F \right\} \right\}, \\ R_{3\rightarrow} = \max \left\{ 0, \min \left\{ 1 + R_{2\rightarrow}, F \right\} \right\}, \\ R_{4\rightarrow} = \max \left\{ 0, \min \left\{ 1 + R_{3\rightarrow}, F \right\} \right\}, \end{cases} \quad (150)$$

with solution $\mathbf{R}_{\rightarrow} = (\frac{3}{5}, 0, 1, 2)$ for $F \geq 2$. The not-shocked banks' payments chain through the ring— B_3 pays B_4 , allowing B_4 to pay $2\theta y = 2$ to B_1 —so B_1 survives and only B_2 is liquidated: The ring is constrained efficient in this state.

Now suppose the shocked banks alternate with the not-shocked ones, say B_1 and B_3 . The clearing vector solves

$$\begin{cases} R_{1\rightarrow} = \max \left\{ 0, \min \left\{ -\frac{7}{5} + R_{4\rightarrow}, F \right\} \right\}, \\ R_{2\rightarrow} = \max \left\{ 0, \min \left\{ 1 + R_{1\rightarrow}, F \right\} \right\}, \\ R_{3\rightarrow} = \max \left\{ 0, \min \left\{ -\frac{7}{5} + R_{2\rightarrow}, F \right\} \right\}, \\ R_{4\rightarrow} = \max \left\{ 0, \min \left\{ 1 + R_{3\rightarrow}, F \right\} \right\}, \end{cases} \quad (151)$$

with solution $\mathbf{R}_{\rightarrow} = (0, 1, 0, 1)$. Each not-shocked bank pays a different shocked bank, so each shocked bank receives only $\theta y = 1 < \frac{7}{5}$ and both are liquidated. The planner instead pools the two not-shocked banks' liquidity, ($2\theta y = 2$), and transfers enough to one shocked bank to save it. The ring is thus efficient for some states (σ) but not for others. Its problem is that the direction of liquidity depends on where the shocked and not-shocked banks happen to lie in the ring: When they alternate, liquidity is divided between the two shocked banks rather than concentrated on one of them.

C Notations

To the extent possible, we use bold face letters for matrices and vectors and use italics for scalars; we use single-arrow subscripts for liabilities from one bank to another and double-arrow subscripts for total liabilities from one to many banks. E.g., $\mathbf{F} = [F_{i \rightarrow j}]_{ij}$ is the matrix of interbank liabilities between individual banks; $\mathbf{F}_{\rightarrow} = [F_{i \rightarrow}]_i$ is the vector of banks' total interbank liabilities, i.e. the vector of row sums of $\mathbf{F}$. We use B_i for individual banks and script letters for sets; $B_i \in \mathcal{B}$ and $i \in \mathcal{B}$ are synonymous. We summarize our notations in Table 2, separating those used in the main text from those used only in extensions or proofs.

Table 2: Notations.

Notation	Meaning	Parametric restriction
y	Long-term real asset value	$y > 0$
ℓ	Size of liquidity shock	$\theta y < \ell < y$
θ	Pledgeable fraction of y	$0 < \theta < 1$
σ_i	Indicator of B_i 's shock	$\sigma_i \in \{0, 1\}$
$\boldsymbol{\sigma} \equiv \{\sigma_i\}_i$	Vector of shocks/Aggregate state	$\boldsymbol{\sigma} \in \{0, 1\}^N$
B_i	i th bank	
$\mathcal{B}$	A set of banks	
$\mathcal{B}^c$	Complement of $\mathcal{B}$	
$\mathcal{L}$	Set of banks that are liquidated	$\mathcal{L} \subset \mathcal{D}$
N	Number of banks	
$S = \sum \sigma_i$	Number of shocked banks	
$L = \mathcal{L} $	Number of liquidated banks	
L^*	Minimum L (Lemma 6)	
$F_{i \rightarrow j}$	B_i 's liability to B_j	$F_{i \rightarrow j} \geq 0$
$\mathbf{F} \equiv [F_{i \rightarrow j}]_{ij}$	Matrix of interbank debts	
$F_{i \Rightarrow} \equiv \sum_{j \neq i} F_{i \rightarrow j}$	B_i 's total interbank liabilities	
F	Each bank's total liabilities $F \equiv F_{i \Rightarrow}$ in a regular network; per-link face value in the star and uniform-complete sections	
$F_{i \Leftarrow} \equiv \sum_{j \neq i} F_{j \rightarrow i}$	B_i 's total interbank claims	
$\mathbf{F}_{\Rightarrow} \equiv \{F_{i \Rightarrow}\}_i$	Vector of each bank's total interbank liabilities	
$\hat{F}_{i \rightarrow j} \equiv F_{i \rightarrow j} / F_{i \Rightarrow}$	B_i 's liability to B_j as a fraction of its total liabilities	$0 \leq \hat{F}_{i \rightarrow j} \leq 1$
$\hat{\mathbf{F}} \equiv [\hat{F}_{i \rightarrow j}]_{ij}$	Matrix of interbank debts	$\sum_i \hat{F}_{i \rightarrow j} = 1$ if $F_{i \Rightarrow} > 0$
$R_{i \rightarrow j}$	B_i 's equilibrium repayment to B_j	$0 \leq R_{i \rightarrow j} \leq F_{i \rightarrow j}$
$R_{i \Rightarrow} \equiv \sum_{j \neq i} R_{i \rightarrow j}$	B_i 's total repayment to other banks	
$R_{i \Leftarrow} \equiv \sum_{j \neq i} R_{j \rightarrow i}$	B_i 's total repayment received from other banks	
$\mathbf{R}_{\Rightarrow} \equiv \{R_{i \Rightarrow}\}_i$	Vector of each bank's total equilibrium repayment	$\mathbf{0} \leq \mathbf{R}_{\Rightarrow} \leq \mathbf{F}_{\Rightarrow}$
$\mathbf{R}_{\Leftarrow} \equiv \{R_{i \Leftarrow}\}_i$	Vector of each bank's total payment received	$\mathbf{R}_{\Leftarrow} = \hat{\mathbf{F}}^\top \mathbf{R}_{\Rightarrow}$
α	Scale of debts used in, e.g., Proposition 2	$\alpha > 0$

Continued on next page

Notations (continued)

Notation	Meaning	Parametric restriction
β	Bottleneck parameter (Definition 6)	
$d_{i \rightarrow j}$	Harmonic distance from B_i to B_j (Definition 5)	$d_{i \rightarrow j} \geq 0$
δ	Connectedness parameter (Definition 4)	$0 \leq \delta \leq 1$
d^{ST}, d^{LT}	Default and salvation radii in Lemma 4 and Proposition 4	
$\beta^{ST}, \beta_{ST}, \beta^{LT}, \beta_{LT}$	Thresholds in Lemma 5 and Proposition 5	
s	Dominance parameter (Definition 9)	$0 < s < 1$
s^*	Threshold in Proposition 6	
t_i	Transfer to B_i in Definition 7	
i^*	Index of highest-ranked liquidated bank in Section 5 and Section 7	
π_i	Permutation of banks keeping B_i fixed	
$[\cdot]^+ = \max\{\cdot, 0\}$	Maximum of variable and zero	
$\lceil \cdot \rceil, \lfloor \cdot \rfloor$	Ceiling and floor functions	

Notations Used Only in Extensions

y^*	Efficient liquidation threshold in Section 8.1	
Δ_i	Efficiency loss if B_i is liquidated in Section 7	
$\mathbf{x} \equiv \{x_i\}_i$	Vector of indicators of banks not being liquidated in Section 7	$\mathbf{x} \in \{0, 1\}^N$
$\hat{\mathbf{t}}, \check{\mathbf{t}}, \hat{\mathbf{x}}, \check{\mathbf{x}}$	Optimizers in Section 7	
π	Ranking of banks for greedy algorithm in Section 7	
y_i	B_i 's long-term real asset value in Section 7	$y_i > 0$
ℓ_i	Size of B_i 's liquidity shock in Section 7	$\theta y_i < \ell_i < y_i$
θ_i	B_i 's chosen pledgeability in Section 8.3	$0 \leq \theta_i \leq 1$
$\theta^a \equiv \ell/y$	Pledgeability needed to meet a shock unaided, Section 8.3	
$c(\cdot)$	Cost of pledgeability in Section 8.3	$c(0) = 0, c', c'' > 0$
p	Probability a bank is shocked in Section 8.3	$0 < p < 1$

Notations Used Only in Proofs

$\mathbf{Q}$	A matrix, usually shorthand for $\hat{\mathbf{F}}^\top$	
$\mathbf{Q}_{\mathcal{B}_1, \mathcal{B}_2} \equiv [Q_{ij}]_{i \in \mathcal{B}_1, j \in \mathcal{B}_2}$	Block matrix with rows in $\mathcal{B}_1$ and columns in $\mathcal{B}_2$	

Continued on next page

Notations (continued)

Notation	Meaning	Parametric restriction
$\tilde{\mathbf{Q}}$	Matrix in Lemma A.8	
$\mathbf{0}, \mathbf{1}$	Vectors of zeros and ones $((0, \dots, 0)$ and $(1, \dots, 1)$)	
$\mathbf{d}_{i \rightarrow \mathcal{B}} \equiv \{d_{i \rightarrow j}\}_{j \in \mathcal{B}}$	Vector of B_i 's harmonic distance to banks in $\mathcal{B}$	
$D_i \equiv F_{i \rightrightarrows} - R_{i \rightrightarrows}$	B_i 's shortfall	
$\mathbf{D} \equiv \mathbf{F}_{\rightrightarrows} - \mathbf{R}_{\rightrightarrows}$	Vector of each bank's shortfall	
$\mathcal{D}$	Set of banks that default	
Φ^α, Ψ^α	Mappings used in Lemma A.6 and Lemma A.1	
$\mathcal{J}^\alpha, \mathcal{H}^\alpha$	Restricted domains of Φ^α and Ψ^α	
$\prod_{i=1}^N X_i = X_1 \times \dots \times X_N$	Cartesian product of sets $X_1, \dots, X_N$	

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